

LESSON PLAN - 16

I PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Akansha
Name of the Practising School : G. S. South Point
Date :
Period : IV
Class : VII
Subject : Mathematics
Name of the lesson : Parallel & intersecting line
Name of the Topic : Introduction to parallel & intersecting line
Duration : 40 mins

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the End Students will be able to-

- define intersecting lines
- Explain parallel lines
- identify parallel and intersecting lines
- differentiate between parallel and intersecting lines by stating two points.

III ENTRY BEHAVIOUR

Que 1: what is a line?

Ans: A line is a straight path that extends endlessly in both directions.

Que 2: what is a line segment?

Ans: A line segment is a part of a line that has two fixed end points.

classmate
Date _____
Page _____

Ques 3: what is a ray?

Ans A ray is a part of line that starts at one point and goes on endlessly in one direction.

Ques 4: can two lines meet each other?

Ans Yes two lines can meet or cross each other at a point.

Ques 5: Do all lines always meet each other or can they also run side by side without touching?

Ans Silence

IV TEACHING AID

Flannel Board representing parallel and intersecting line.

V ANNOUNCEMENT OF THE TOPIC

So, Students today we are going to study about Introduction to parallel and intersecting lines

INSTRUCTIONAL PROCEDURE

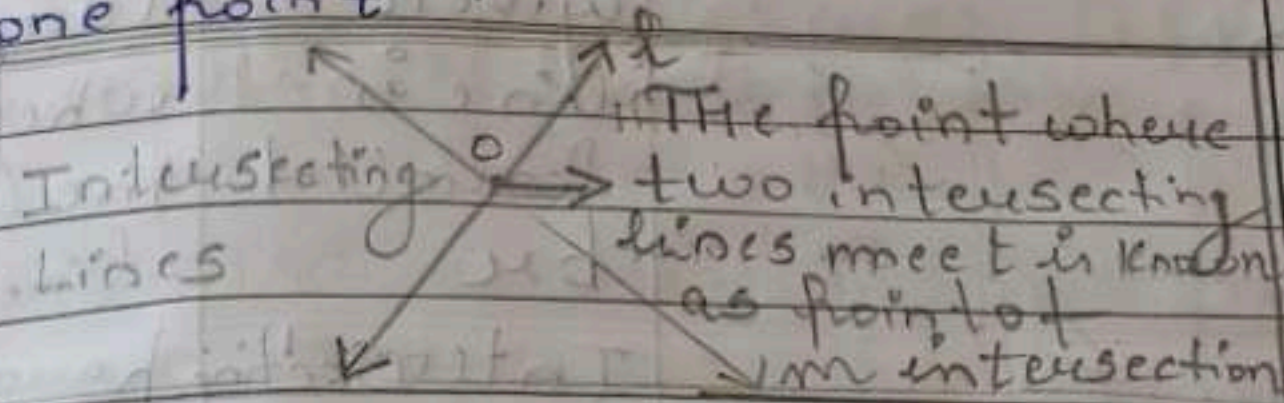
Learning Experiences	Behavioural Objectives	Evaluation
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Intersecting Lines

Intersecting lines

Intersecting lines are two or more lines that meet or cross at exactly one point.

Intersecting lines are lines that cross each other at one point.



Name: _____
the point where two lines meet.

Parallel Lines

Parallel lines

Parallel lines are a pair of lines that lie on the same plane, and do not meet however far we extend them at both the ends.

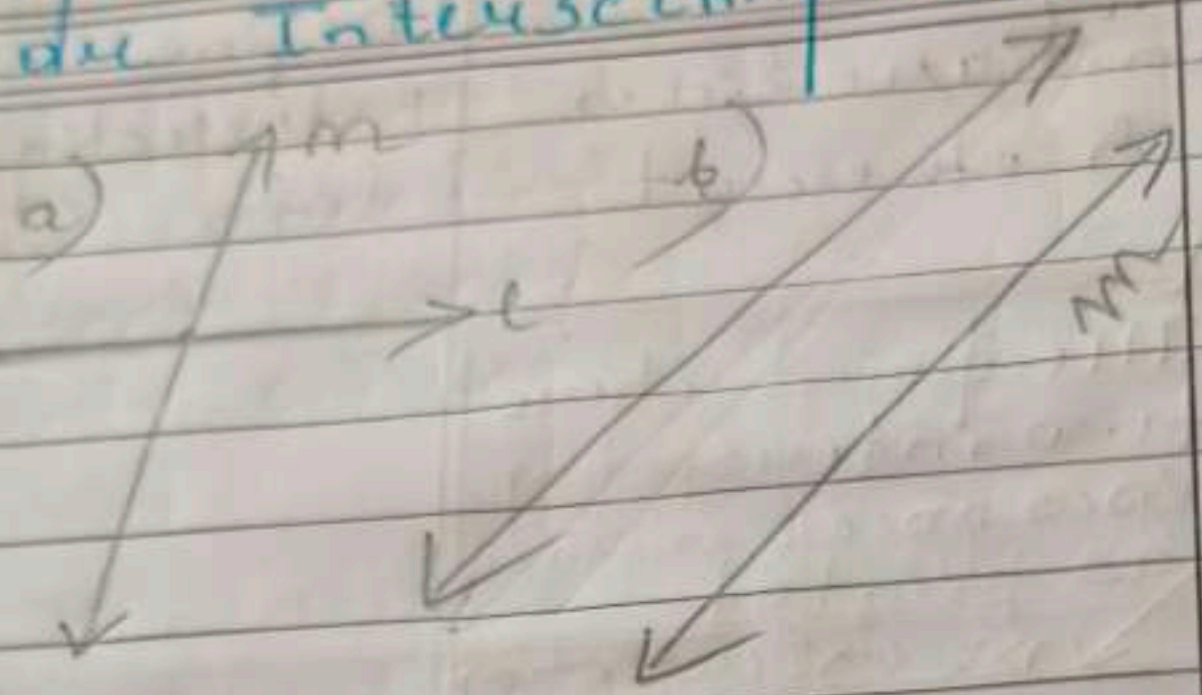
Parallel lines are lines that run side by side and never meet, no matter how far they are extended.

Eg: 1) The two long lines on railway tracks are parallel.

2) The opposite sides of a rectangle are parallel.

Students Explain parallel lines.

Explain parallel lines

content	Learning Experiences	Behavioural objectives	Evaluation
PARALLEL LINES line l is $\parallel$ to line m Identify whether the given line is parallel or intersecting lines.  In Fig a) line l intersects line m at a single point, and the definition of intersecting line is two lines or more lines that meet each other at exactly one point is known as intersecting line. So, Fig a is intersecting line. In Fig b) The lines in Fig b lie on the same plane and do not meet however far they are extended. So, the	Identify whether the given line is parallel or intersecting lines. Identify whether the given line is parallel or intersecting lines.	Students identify whether the given line is parallel or intersecting lines. Identify whether the given lines are parallel or intersecting lines.	differentiate b/w parallel and intersecting lines

Learning
Experiences

line is parallel. So, line m

Behavioural
objectives

Evaluation

Differentiate between parallel and intersecting lines

• Parallel lines never meet
even if extended, whereas
intersecting lines meet or
cross at one point.

• Parallel lines do not form
angles with each other,
whereas intersecting lines
form angles at the
point they meet.

• Parallel lines always
remain the same
distance apart, whereas
intersecting lines do not
maintain equal distance.

• Example of parallel lines:
the two rails of a train
track are an example of
parallel lines, whereas
example of intersecting lines:
the lines that form the
letter (X) or (+)
are intersecting lines.

Students
differentiate
between

parallel
and
intersecting
lines.

write
the
difference
between

parallel
and
intersecting
lines.

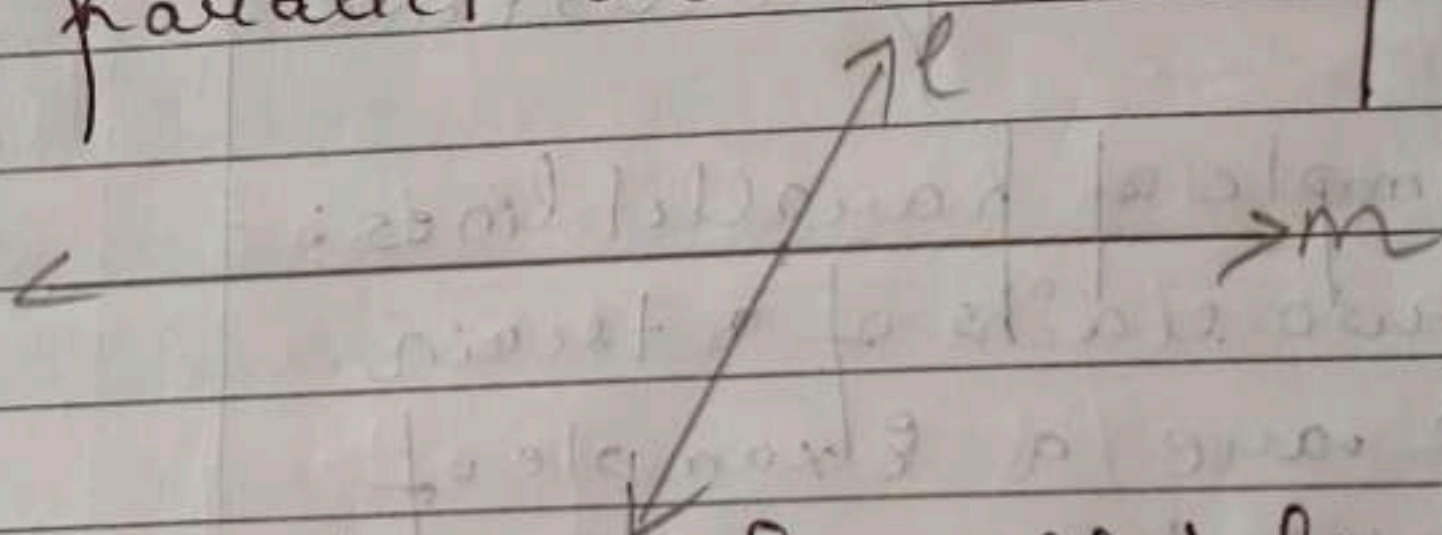
VII SUMMARY

- Intersecting lines are two or more lines that meet or cross at exactly one point.
- Parallel lines are two lines that lie on the same plane and never meet, even when extended.
- Difference between parallel and intersecting lines, parallel lines never meet even if they are extended whereas intersecting lines meet exactly at one point.

VIII EVALUATION

Que 1: Define intersecting lines? Give one example.

Que 2: Look at the figure and tell whether the given pair of lines are parallel or intersecting.



Que 3: What are parallel lines?

Que 4: Name the point where two intersecting lines meet.

Ans

Signature of the
Teacher Trainee

Signature of the
Guide Teacher

LESSON PLAN - 17

I PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Arkantha
Name of the Practising School : G. S. South Point
Date :
Period : IV
Class : VII
Subject : Mathematics
Name of the Lesson : Parallel & Intersecting lines
Name of the topic : Vertically opposite angles and linear pair
Duration : 40 mins

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the lesson, student will be able to

- define vertically opposite angles and a linear pair
- describe how two intersecting lines form four angles and the opposite angles among them are vertically opposite angles and are equal.
- explain that in a linear pair the two adjacent angles always add up to 180°
- verify whether a given pair of angles represent vertically opposite angles or a linear pair.

III TEACHING AID

Flashcard representing Vertically opposite angle and Linear pair.

IV ENTRY BEHAVIOUR

Ques: what is an angle?

Ans: An angle is the figure formed by two rays meeting at a common point.

Ques: what are the names of basic types of angles?

Ans: Acute, right, obtuse, straight, reflex angles.

Ques: what are intersecting lines?

Ans: Intersecting lines are lines that cross each other at one point.

Ques: what do you call two angles that are next to each other?

Ans: Two angles that share a common arm and vertex are called adjacent angles.

Ques: when two lines intersect, they form four angles. Some of these angles are opposite each other and some are next to each other forming straight line. What do you think these opposite angle and straight line angles are called?

Ans: Silence

V ANNOUNCEMENT OF THE TOPIC

So, student today we are going to study about Vertically opposite angle and Linear pair.

VI INSTR

content

linear pair

linear

A pair of adjacent angles that form a straight line is called a linear pair. Each angle is called a linear angle.

Vertically opposite angle

INSTRUCTIONAL PROCEDURE

Meaning Experience	Behavioural objectives	Evaluation
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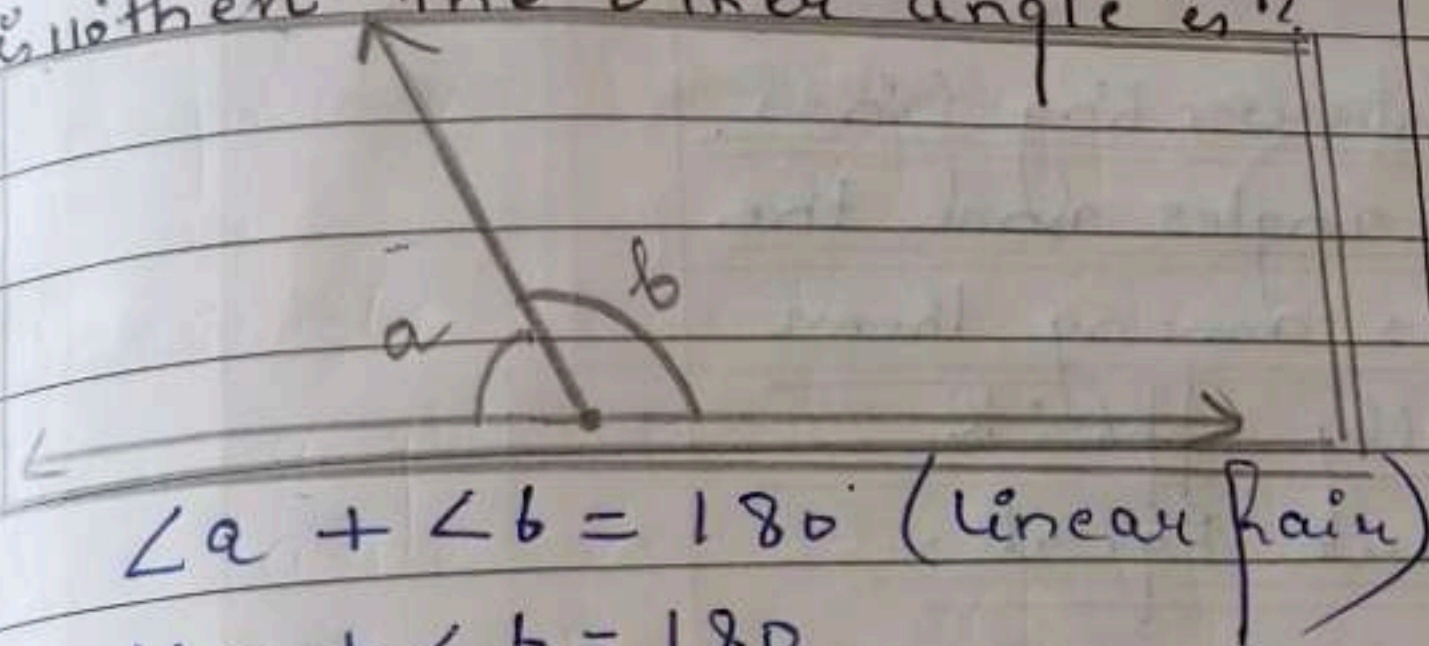
Linear pair

A pair of two or more adjacent angles that are next to each other on a straight line, their sum is 180° .

When two angles are next to each other and together they make a straight line, they form a linear pair, because a straight line is 180° , both angles add up to 180° .

Example:

If two adjacent angles form a straight line and one angle is 110° , then the other angle is?



$$110 + \angle b = 180$$

$$\angle b = 180 - 110$$

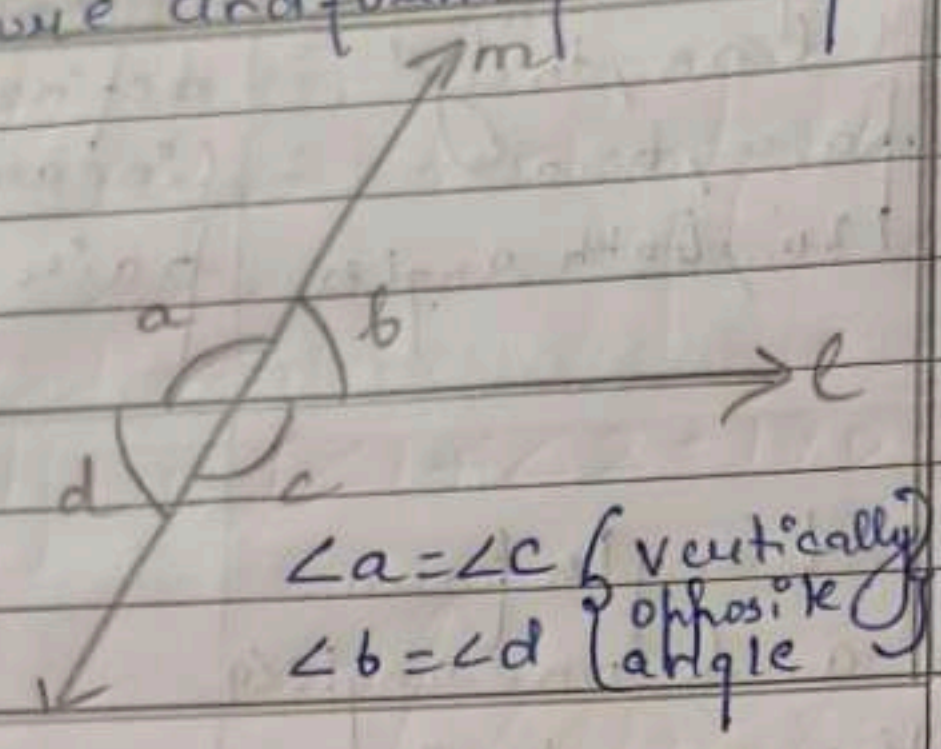
$$\boxed{\angle b = 70^\circ}$$

Vertically opposite angles

Vertically opposite angles are the pair of angles formed opposite to each other when

Students	Explain
define	Vertically
Vertically	opposite
opposite	angle.
angle	

content meaning
 Experience
 two lines intersect. these
 angles are always equal
 in measure.
 opposite angles formed by
 two intersecting lines
 are known as vertically
 opposite angle. And a pair
 of vertically opposite
 angles is always equal in
 measure and form four angles.



Behavioural objectives

Evaluation

Define vertically opposite angles.

How many angles do they form?

How two intersecting lines form four angles and the opposite angles among them are vertically opposite angles and are equal.
 When two straight line intersect, they form four angles. The pairs of angles that are directly opposite each other are called vertically opposite angle, and they are always equal. This relationship

Students describe how two intersecting lines form four angles and

Evaluation

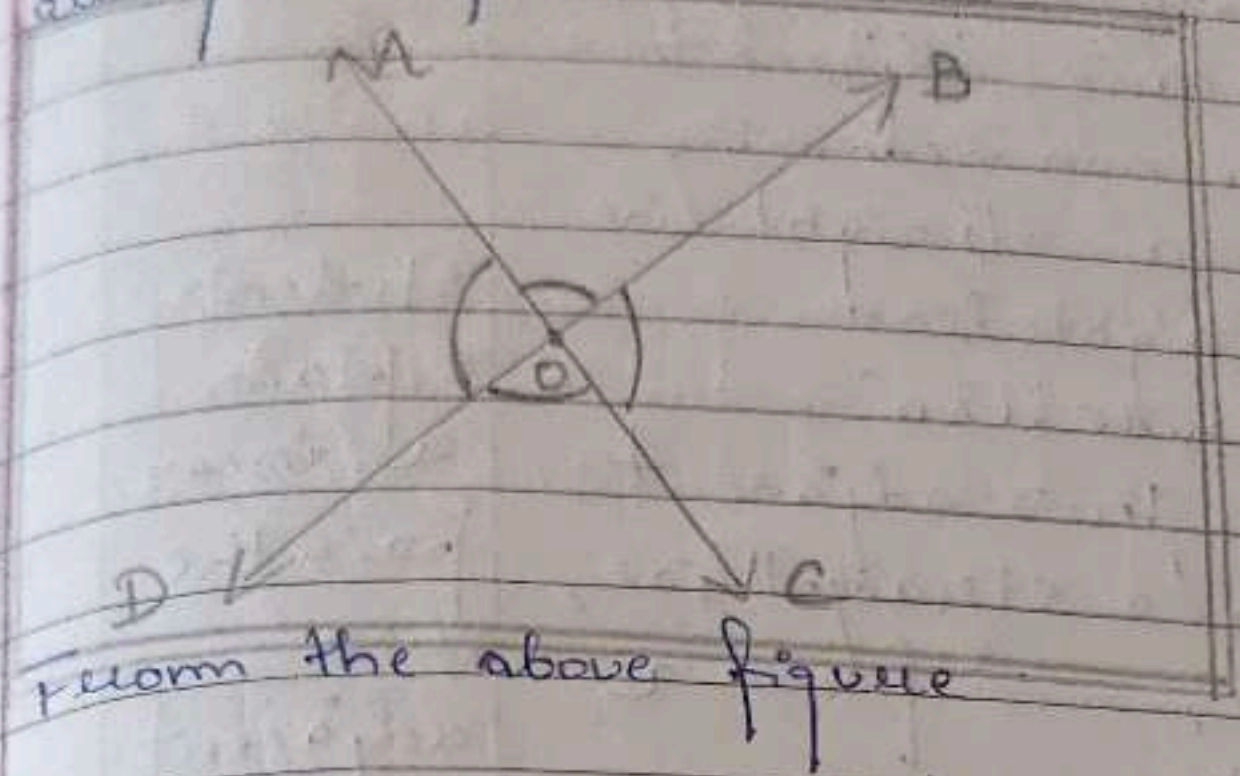
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many
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form?

Learning

Experiences

can be proved using the properties of a linear pair of angles, which are adjacent angles that form a straight line and always add up to 180.



From the above figure

$$\angle AOB + \angle BOC = 180 \text{ (Linear pair)} \quad \text{--- (1)}$$

$$\angle BOC + \angle COD = 180 \text{ (Linear pair)} \quad \text{--- (2)}$$

From Eq (1) and (2)

$$\angle AOB + \angle BOC = \angle BOC + \angle COD$$

$$\angle AOB + \cancel{\angle BOC} - \cancel{\angle BOC} = \angle COD$$

$$\angle AOB = \angle COD \text{ (vertically opposite angle)}$$

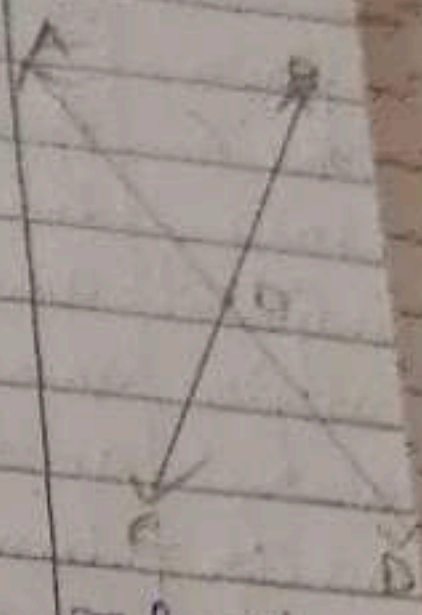
$$\text{Similarly } \angle DOA = \angle BOC \text{ (vertically opposite angle)}$$

Hence proved that $\angle AOB = \angle COD$
 $\angle DOA = \angle BOC$ are vertically opposite angle and are Equal.

Behavioural objectives

the opposite angle among them are V.O.A and are Equal

Evaluation



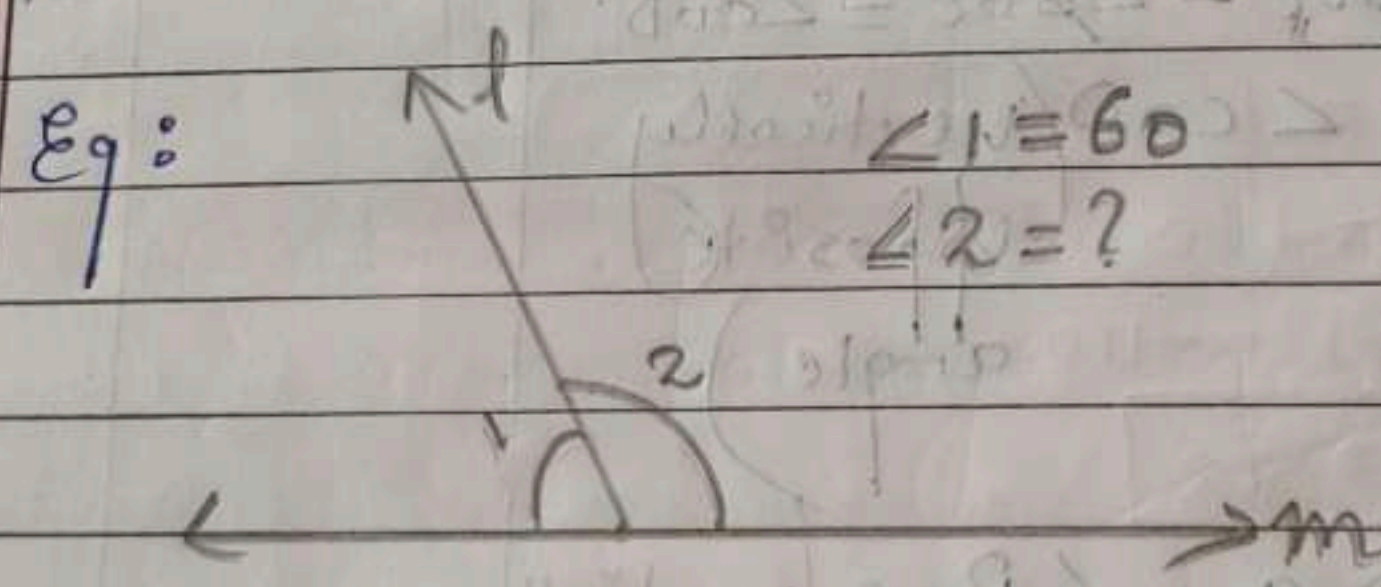
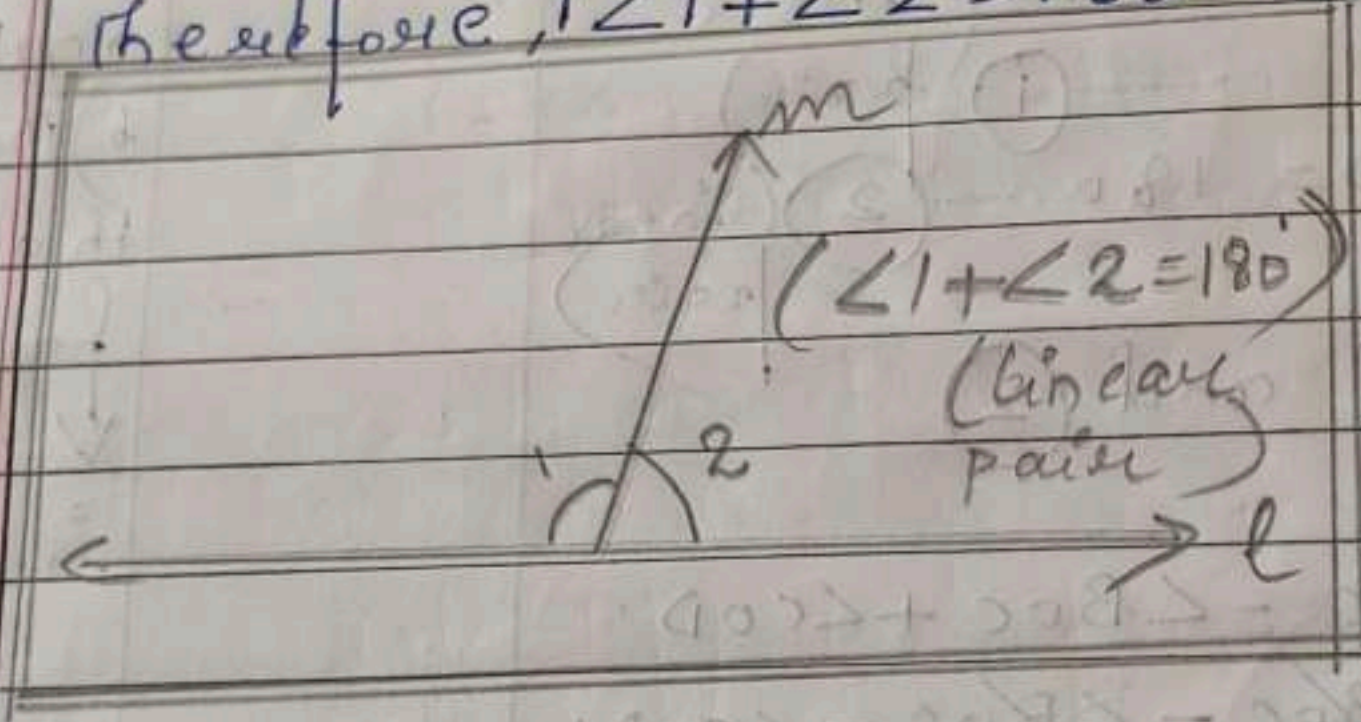
Identify the pairs of vertically opposite angle.
 b) Prove that each pair of V.O.A is Equal.

content meaning Experiences
Explain in a linear pair the two adjacent angles always add up to 180°

A linear pair is formed when two angles are adjacent and their non-common arms form a straight line. always measure 180°

Sp, when two adjacent angles lie on a straight line, their non-common form a straight angle.

Therefore, $\angle 1 + \angle 2 = 180^\circ$



$\angle 1 + \angle 2 = 180^\circ$ (Linear pair)
 $60 + \angle 2 = 180$
 $\angle 2 = 180 - 60$
 $\angle 2 = 120$

$\angle 1 + \angle 2 = 180$; $120 + 60 = 180$

Behavioural objective

Evaluation

Students Explain a linear pair the two adjacent angles always add up to 180°

Are the angles in a linear pair always Supplementary?

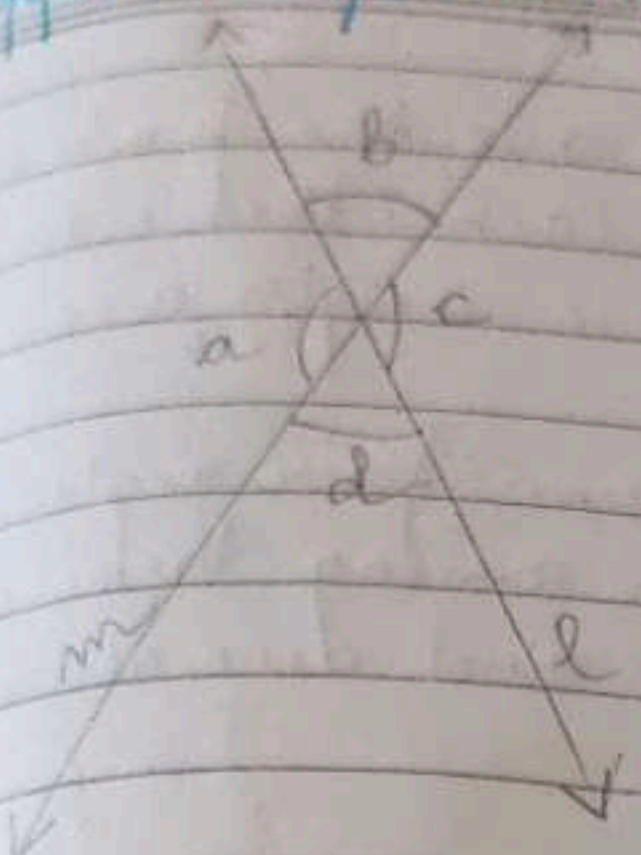
content
Verify whether a given pair of adjacent vertically opposite angles on a linear pair.

Evaluation

Learning

Experiences

Verify whether a given pair of angles represent vertically opposite angles on a linear pair.



• $\angle a$ and $\angle c$ are vertically opposite angles and $\angle b$ and $\angle d$ are vertically opposite angle.

$\therefore \angle a = \angle c$ (vertically opposite angle)
 $\angle b = \angle d$

• $\angle a$ and $\angle b$ form a linear pair (they are adjacent and lie on a straight line).

$\angle b$ and $\angle c$ form a linear pair.

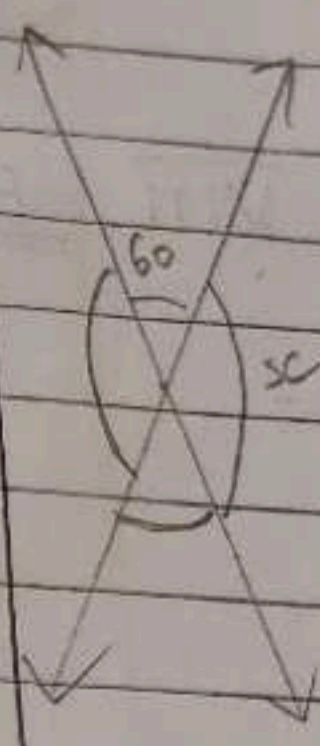
$\angle c$ and $\angle d$, $\angle d$ and $\angle a$ form a linear pair.

Each pair of adjacent angles ($\angle a$ and $\angle b$, $\angle b$ and $\angle c$, $\angle c$ and $\angle d$, $\angle d$ and $\angle a$) are linear pairs because they are adjacent and supplementary.

Behavioural objectives

Evaluation

Find the value of x .



Students verify whether a given pair of angles represent vertically opposite angles on a linear pair.

VII SUMMARY

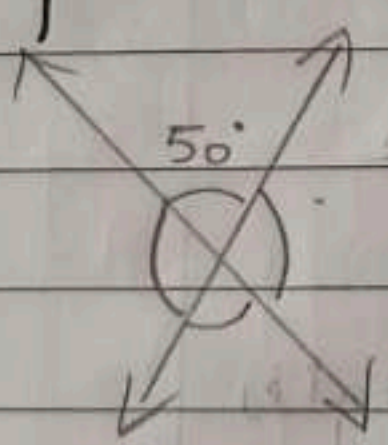
- A pair of two or more adjacent angles that are next to each other on a straight line, their sum is 180° is said to be linear pair.
- Vertically opposite angles are the pair of angles formed opposite to each other when two lines intersect. These angles are always equal in measure.
- Two intersecting lines form four angles and the opposite angles formed among them are vertically opposite angle and are always equal.

VIII EVALUATION

Que 1: Can two straight lines intersect at more than one point?

Que 2: Draw two intersecting lines and label the four angles formed. Identify the pairs of vertically opposite angles.

Que 3: Why are vertically opposite angles always equal?

Que 4:  Find the measure of the other vertically opposite angle.

Ans

Signature of the
Teacher Trainee

Signature of the
Lykide Teacher

LESSON PLAN - 18

PRELIMINARY INFORMATION

Name of the Teacher Trainee: S. Akasha
Name of the Practising School: GSS South Point
Date: :
Period: : VII
Class: : VII
Subject: : Mathematics
Name of the Lesson: : Parallel and intersecting lines
Name of the topic: : Perpendicular lines
Duration of the Period: : 40 min.

SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class student will be able to

- i) define perpendicular lines
- ii) explain with examples how two lines form a right angle when they are perpendicular.
- iii) identify perpendicular lines in a given diagrams.
- iv) differentiate between parallel lines and perpendicular lines based on their properties.

TEACHING AID

Flash card representing perpendicular lines

IV ENTRY BEHAVIOUR

ques: when two straight line meet, what do we call that meeting point?

Ans: Intersection point

ques: can two lines intersect in more than one way?

Ans: Yes, they can intersect in different ways

ques: Do intersecting lines form different types of angles or the same type of angle each time?

Ans: They form different types of angles

ques: Is it possible for two lines to intersect in a very specific positional arrangement?

Ans: Yes it is possible

ques: what do we call the pair of lines that intersect in this specific positional arrangement?

Ans: Perpendicular lines

V ANNOUNCEMENT OF THE TOPIC

So, students today we are going to study about Perpendicular lines

VI Intro

content

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Example

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INSTRUCTIONAL PROCEDURE

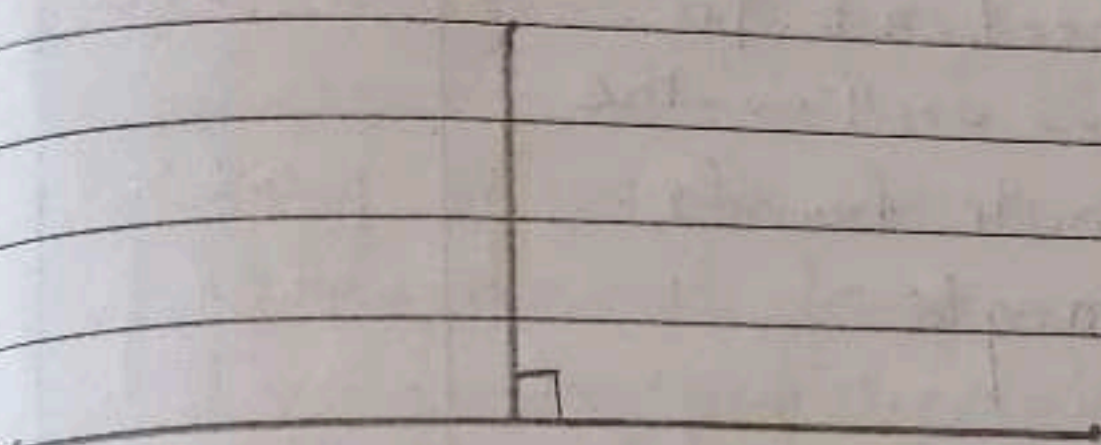
Learning

Experience

Perpendicular Lines

Two lines are called perpendicular lines if they intersect each other and form a right angle (90°) at the point of intersection.

Perpendicular lines are lines that cross each other in such a way that the angle formed between them is a right angle.



Explain with Example how two lines form a right angle

Two lines form a right angle when they intersect at 90° angle. This is also known as a right angle, which is represented by a small square at the intersection point.

Behavioral objectives

Evaluation

Students

define

perpendicular lines

Define

perpendicular lines

students

Explain

how two

lines form

a right

angle

Learning Experiences

Example: The corner of a room

- Imagine the corner where a wall meets the floor in a rectangular room.
- The edge of the wall and the edge of the floor are two lines that intersect at the corner.
- They meet at a 90° angle forming a right angle at the corner.

- * Floor tiles - The lines on many tile meet at 90°
- * Clock at 3:00 or 9:00 - The hour and minute hands form a right angle.

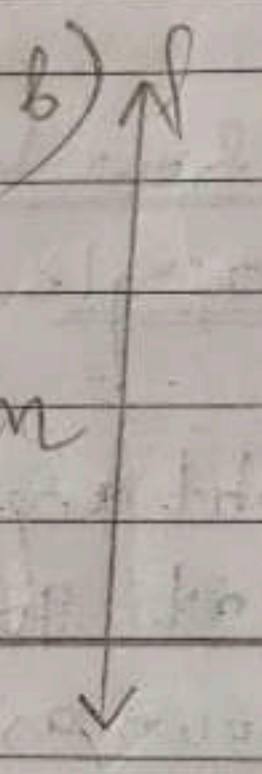
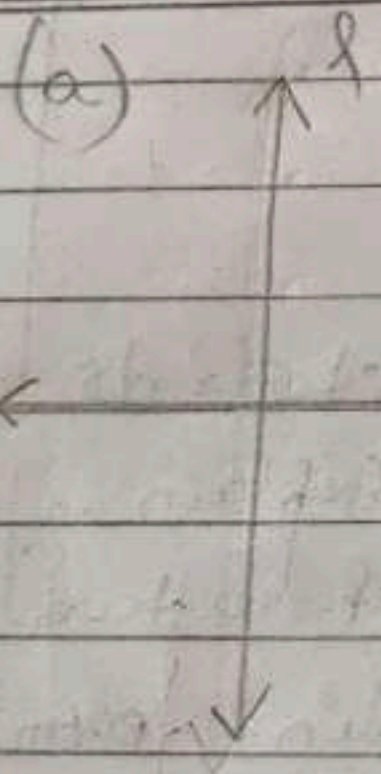
Behavioural objectives

Evaluation

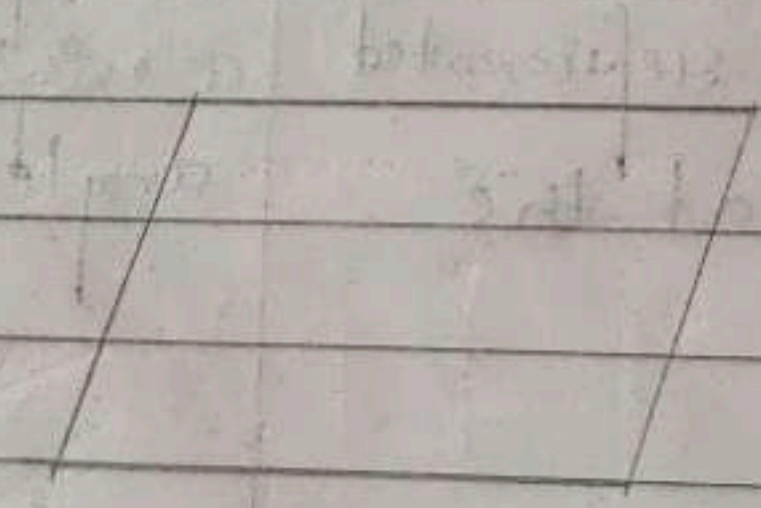
Give some examples of perpendicular lines.

Identify Perpendicular lines

IDENTIFY PERPENDICULAR LINES

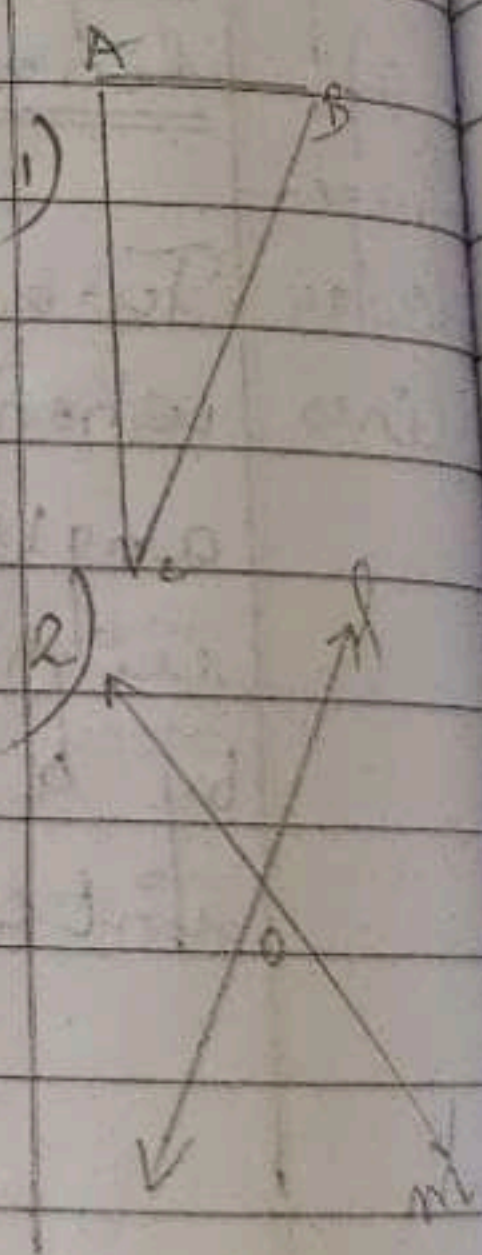


c)



Students identify perpendicular lines

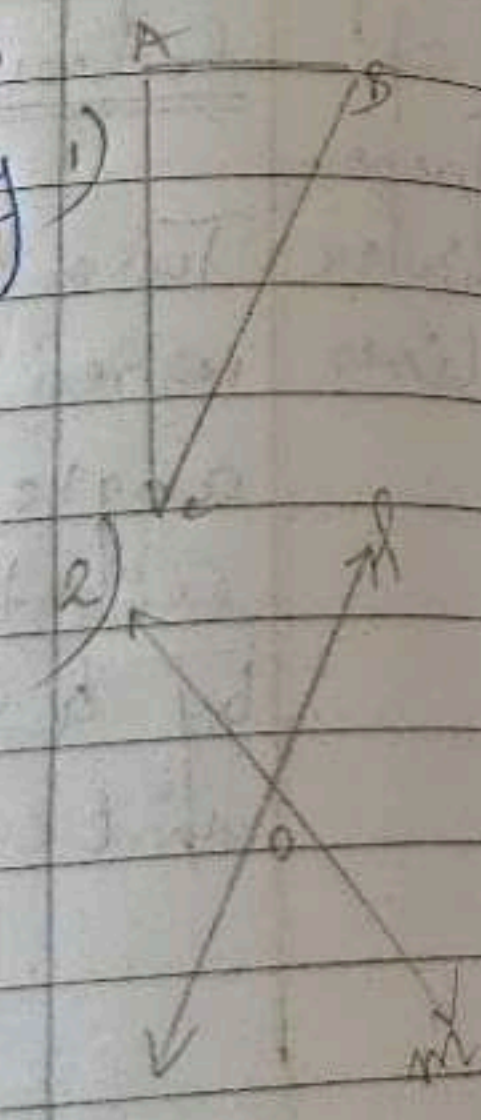
Identify perpendicular line



1) Evaluation

Give some examples of perpendicular lines.

Identify perpendicular line



Meaning of Perpendicular

In Fig (a) two lines (l and m) intersect at a 90° angle. therefore it is perpendicular line.
In Fig (b) two lines (l and m) intersect each other at 90° forming right angle, therefore it is perpendicular line.
In Fig (c) The lines are not perpendicular, because they do not form a right angle (90°)

Differentiate between Parallel and perpendicular lines.

Differentiate between parallel and perpendicular lines

Parallel lines - They never meet, Even if extended. They are always the same distance apart. They do not form any angle because they do not intersect. Example Railway tracks, Edges of a book etc.
Perpendicular lines are the lines that intersect each other at right angle (90°). The angle formed at the intersection is exactly 90° .
Examples: The corner of a square or rectangle, the hands of a clock at 3:00 or 9:00

Behavioural objectives

Evaluation

Students differentiate between parallel and perpendicular lines

differentiate between parallel and perpendicular lines

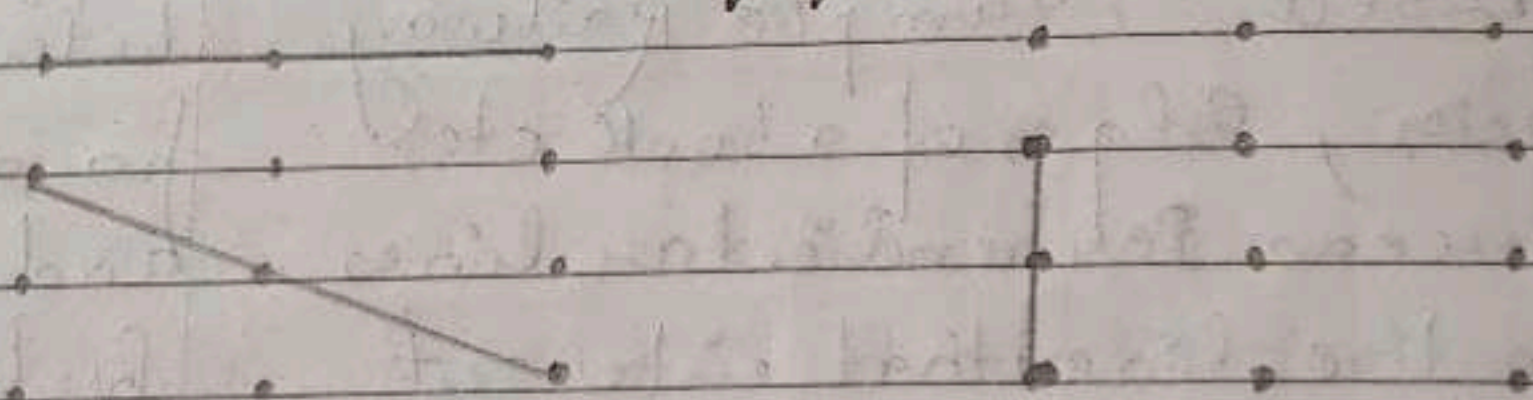
VII SUMMARY

- Perpendicular lines are lines that cross each other in such a way that the angle formed between them is a perfect right angle.
- Examples of perpendicular lines: corners of a room, floor tiles, clock at 3:00 or 9:00 etc.

• Parallel lines are lines that run side by side without crossing each other. They do not intersect at any point and remain equal distance everywhere.

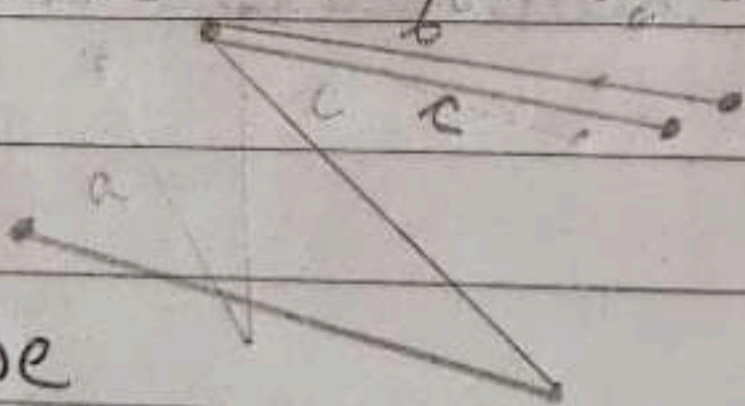
VIII EVALUATION

Que 1: Draw some lines perpendicular to the lines given on the dot paper in the below figure.



Que 2: Draw a pair of intersecting lines such that all four angles are equal.

Que 3: In the below figure which line is parallel to line a — line b, or line c.



Answer
Signature of the
Teacher Trainee

Signature of
the Guide
Teacher

LESSON PLAN - 19

PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. AKANSHA
Name of the Practising School : G. S. South Point
Date :
Period : VII
Class : VII
Subject : Mathematics
Name of the Lesson : Parallel and intersecting lines
Name of the topic : Transversal and corresponding angles
Duration of the period : 40 minutes

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class student will be able to -

- i) define a transversal
- ii) describe the term corresponding angles
- iii) state the property of corresponding angles when a transversal cuts parallel lines
- iv) construct and label a transversal and the corresponding angle pairs
- v) solve simple angle based problem involving corresponding angles

III TEACHING AID

Cardboard representing how a transversal

cuts two lines and forms corresponding angles, enabling students to visualize the concept clearly.

IV ENTRY BEHAVIOUR

Que 1: Define Intersecting line.

Ans: Intersecting lines are lines that meet each other at a single point.

Que 2: If two lines never meet when extended and same distance apart, what are these lines called?

Ans: Parallel lines

Que 3: What is the angle formed by perpendicular lines?

Ans: Right angle i.e. 90°

Que 4: Give some Example of Perpendicular line.

Ans: Corner of a room, clock at 3:00 or 9:00.

Que 5: If I draw a line that cuts parallel lines at different point.

Do you know what this line is called and do you know what are the angles formed?

Ans

V ANNOUNCEMENT OF THE TOPIC

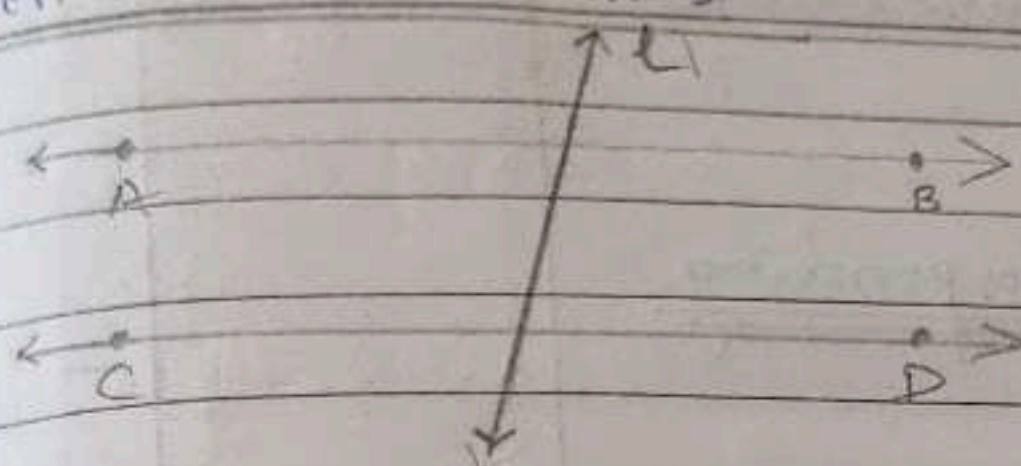
So, students today we are going to study about "Transversal and corresponding angles".

INSTRUCTIONAL PROCEDURE

Learning
Experiences

Transversal

A transversal is a line that cuts or intersects two or more lines at different points.
When a line passes across other lines and forms angles at each point of intersection, it is called a transversal. It creates eight angles when it cuts two lines.



Here the line l represents the transversal.

Corresponding Angles

Corresponding angles are pairs of angles that are in the same relative position when a transversal cuts two lines.

When a transversal intersects two lines, angles that occupy matching position at each intersection such as both above the line, below the line, or both on the same side of the transversal are called corresponding angles.

Behavioural
Objectives

Evaluation

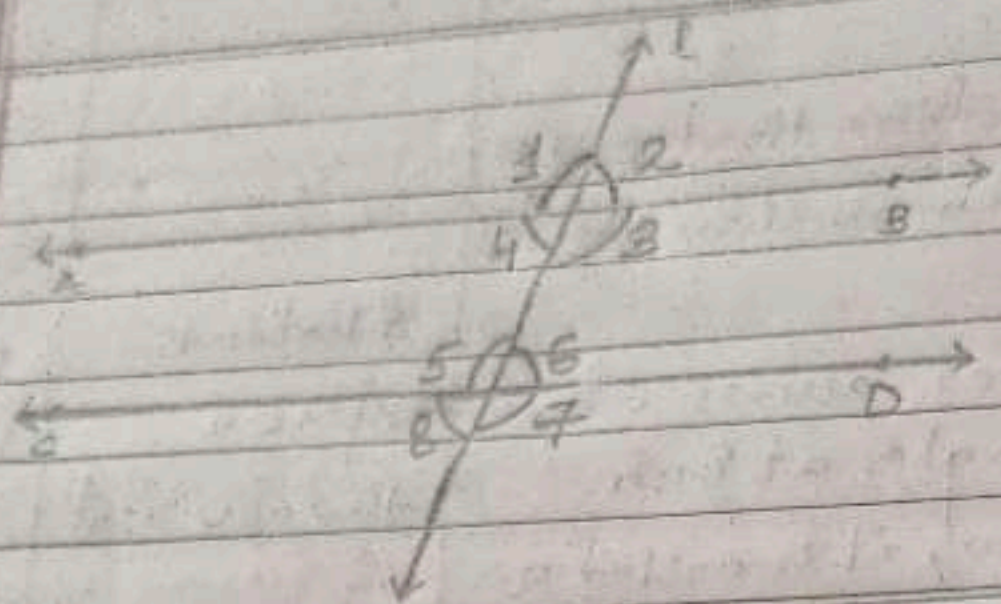
Students
define a
transversal

Define
transversal

Students
describe
the term
corresponding
angle.

What is
corresponding
angle?

Learning Experience
 ding angle. they are Equal when the two lines are parallel.



The angles $\angle 1$ and $\angle 5$, $\angle 2$ & $\angle 6$, $\angle 3$ & $\angle 7$, $\angle 4$ & $\angle 8$ are called corresponding angles.

Property of Corresponding Angles

when a transversal cuts two parallel lines, each pair of corresponding angles are Equal. A transversal create eight angles when it cuts two lines. Among these angles there are four pairs of corresponding angles. These corresponding angle pairs will always be Equal in measure.

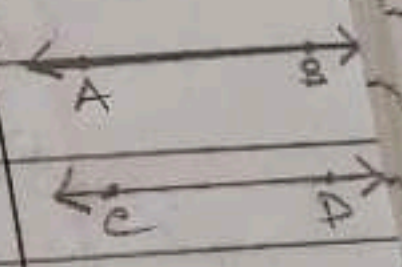
Steps to construct two parallel lines:
 • Draw two straight horizontal lines using a scale.
 • Name them as l_1 and l_2 .

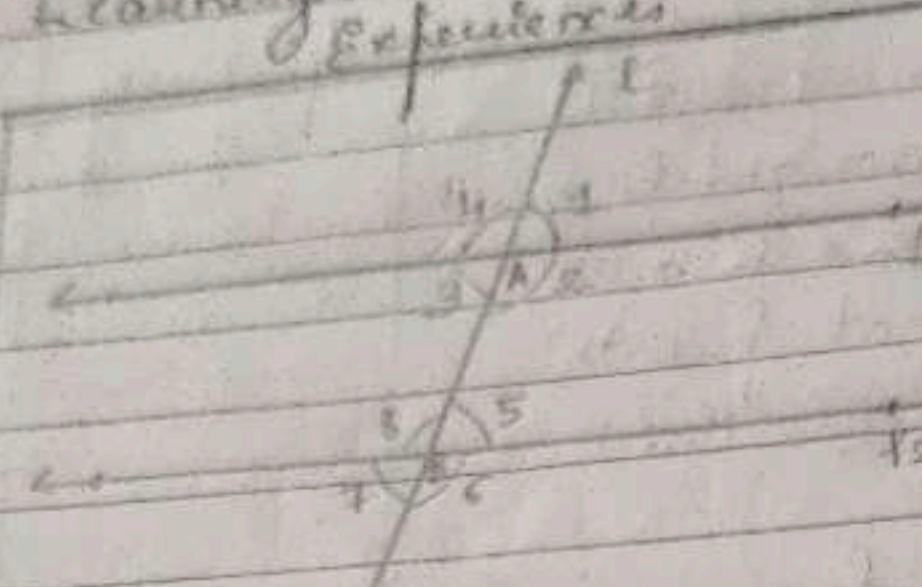
Behavioural objectives

Evaluation

Briefly describe about corresponding angle.

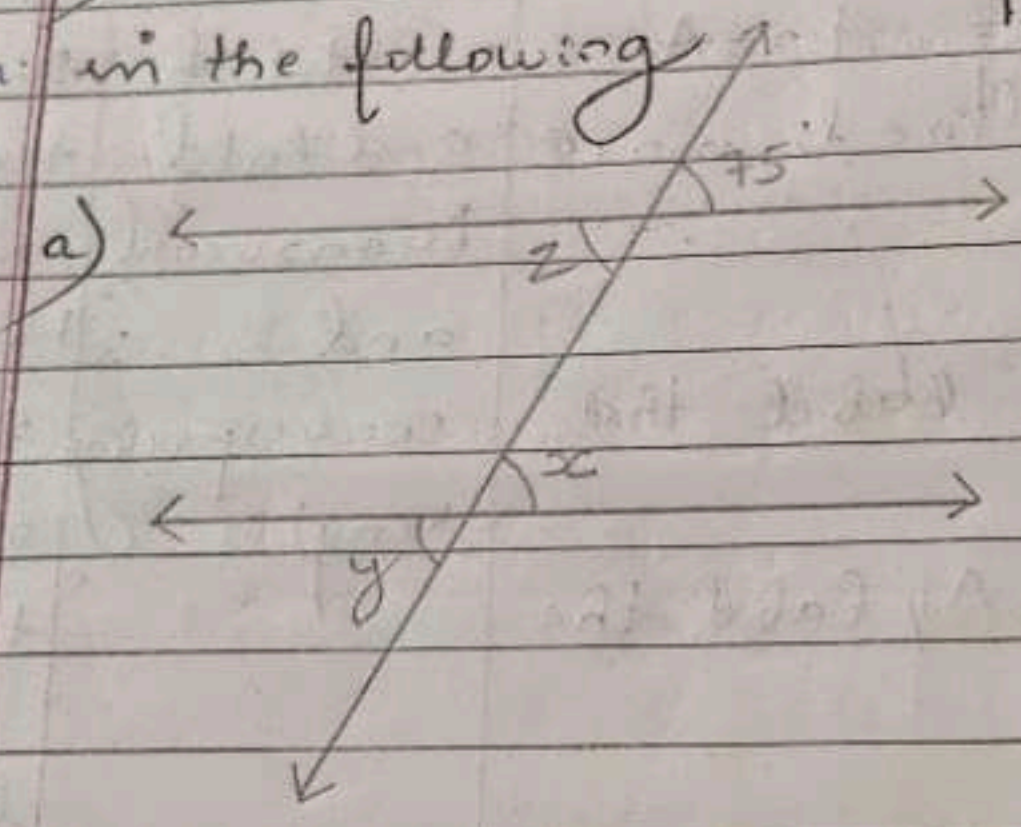
Explain the property of corresponding angles.

Content	Learning Experience	Behavioural Objectives	Evaluation
	Step 2: Draw a transversal • Draw another straight line cutting both the line l_1 and l_2 at two different points. • This slanting line is the transversal • Label it as line t.		
	Step 3: Mark the points of intersection • where the transversal cut line l_1, mark the point as A. • where it cut line l_2 mark the point as B.	Student construct and label transversal and corresponding angles.	Draw a transversal and label the corresponding angles for the given pair of lines.
	Step 4: Form and label the Angles • At intersection A, label the four angles $\angle 1, \angle 2, \angle 3, \angle 4$ • At intersection B label the four angles $\angle 5, \angle 6, \angle 7, \angle 8$		
	Step 5: Identify corresponding angle pairs Now match angles that are in the same relative position at both intersection. The four corresponding pairs are • $\angle 1$ and $\angle 5$ • $\angle 2$ and $\angle 6$ • $\angle 3$ and $\angle 7$ • $\angle 4$ and $\angle 8$		

Content	Learning Experiences	Behavioural objectives	Evaluation
			Construct neat and label diagram containing transversal and corresponding angles

Simple angle based problem

1) Find out the unknown angles in the following



According to the property of corresponding angles

$x = 75^\circ$

Now using the property of vertically opposite angle

$z = 75^\circ$

$y = x = 75^\circ$

$y = 75^\circ$

Evaluation

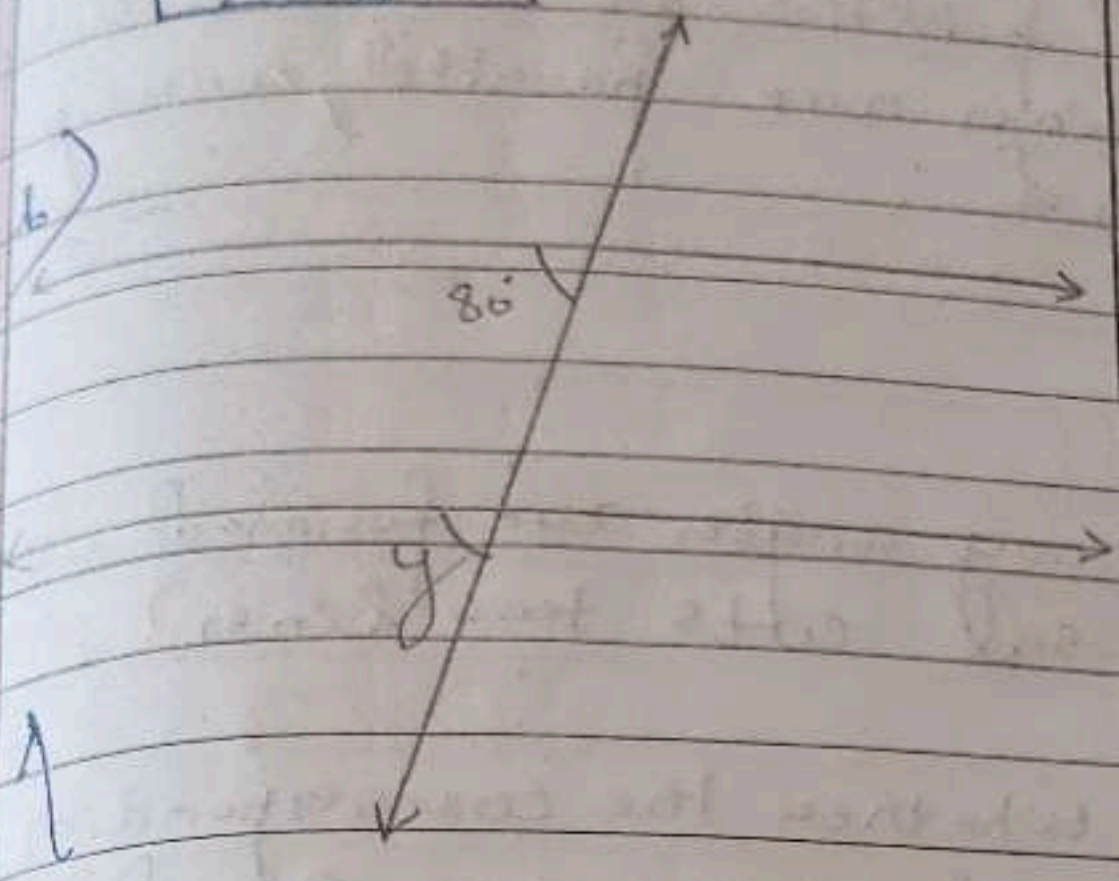
construct neat and label diagram containing transversal and corresponding angle

Learning Experience

Now again using the property of corresponding angles

$$z = y = 75^\circ$$

$$z = 75^\circ$$



According to the property of corresponding angles

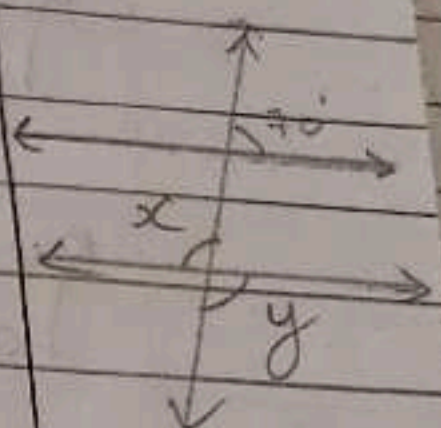
$$y = 80^\circ$$

Behavioural objectives

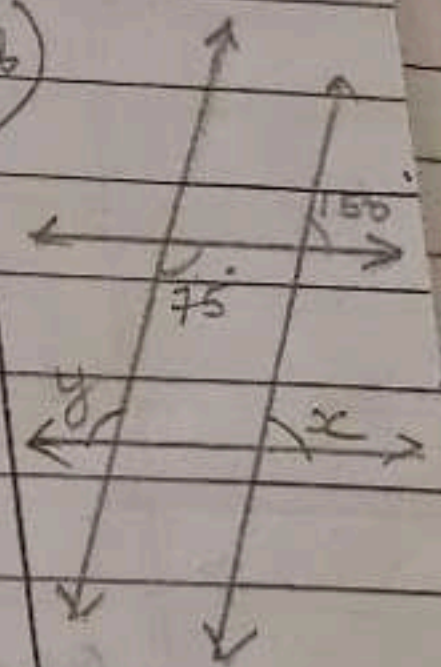
Evaluation

Find out the unknown angles in the following:

a)



b)



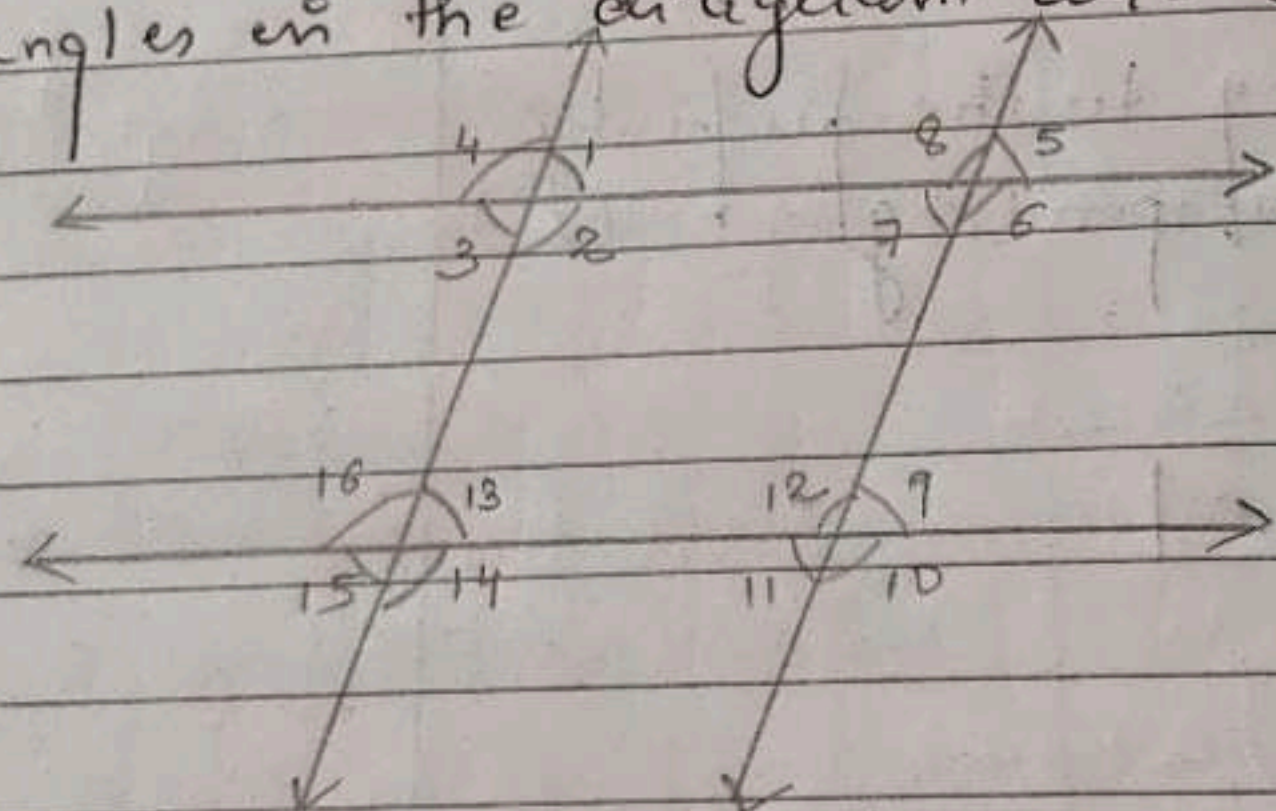
VII SUMMARY

- A transversal is a line that cuts two or more lines.
- when a transversal cuts two lines, eight angles are formed.
- corresponding angles are angle in the same relative position at each intersection.
- when two lines are parallel, corresponding angles are equal.

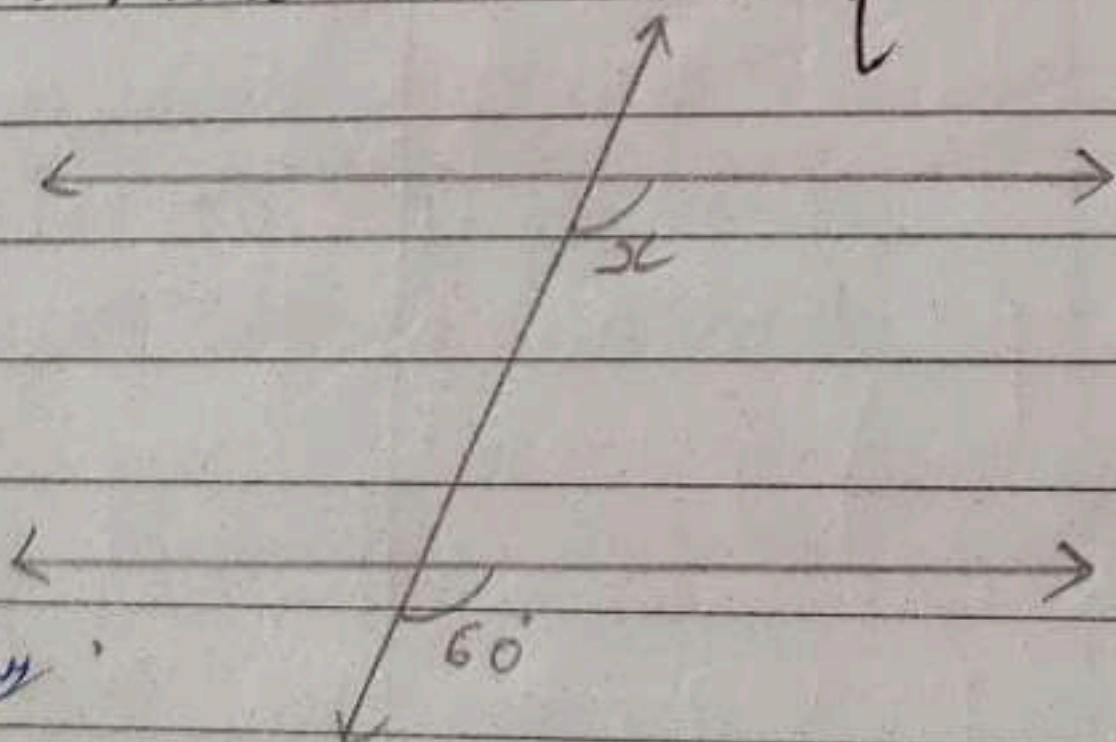
VIII EVALUATION

Que 1: How many angles are formed when a transversal cuts two lines?

Que 2: Check whether the corresponding angles in the diagram are equal.



Que 3: Find the value of x .



Ans:

Signature of the
Teacher Trainee

Signature of the
Guide Teacher

LESSON PLAN - 20

I. PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. AKANISHA
Name of the Practising School : GSS South Point
Date :
Period : VI/V
class : VII
subject : Mathematics
Name of the Lesson : Parallel and intersecting lines
Name of the topic : Alternate angles
Duration of the period : 40 mins

II. SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class student will be able to

- i) define the term alternate angles.
- ii) enlist the types of alternate angles.
- iii) describe alternate interior angles.
- iv) explain alternate exterior angles.
- v) identify pairs of alternate interior angles and exterior angles.
- vi) state the property of alternate angles.
- vii) apply the property of alternate angles to solve simple numerical problems.

III. TEACHING AID

Cardboard representing alternate interior angles

IV ENTRY BEHAVIOUR

ques: what is a transversal?

Ans: A transversal is a line that intersects two or more lines at different points

ques: How many angles are formed when a transversal cuts two lines?

Ans: Eight angles are formed.

ques: Name any two pairs of vertically opposite angles formed when lines intersect.

Ans: $\angle 1$ & $\angle 3$, or $\angle 2$ or $\angle 4$

ques: what are corresponding angles when a transversal intersects two lines?

Ans: Corresponding angles are angles that are in the same relative position at each intersection of the transversal with two lines

ques: If a line cuts two parallel lines, many new angles are formed. Can anyone tell what we call the angle pairs that lie on opposite sides of the transversal but inside the parallel lines?

Ans: Silence

V ANNOUNCEMENT OF THE TOPIC

So, student today we are going to study about "Alternate angles".

VI INST

content: head

Alternate angles

Alter angle sides cuts

with

line

then

angle

Types of Alternate angle

Types of

Alternate

angle

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INSTRUCTIONAL PROCEDURE

Content	Meaning Experiences	Behavioural objectives	Evaluation
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<u>Alternate angles</u>	<u>Alternate angles</u> Alternate angles are the pair of angle that lie on the opposite sides of a transversal when it cuts two lines. when a transversal (a line that cuts two lines) passes through parallel lines, many angles are formed.	Students define the term alternate angle.	
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Types of Alternate Angle

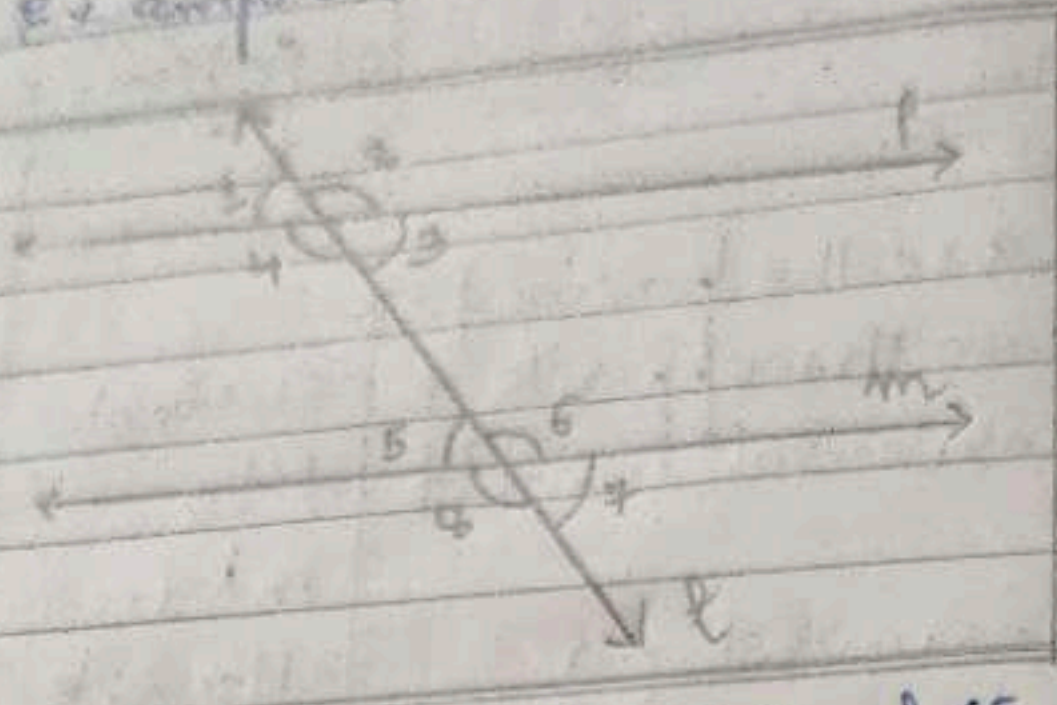
<u>Types of Alternate angle</u>	There are two types of alternate angles: 1) Alternate interior Angles 2) Alternate Exterior Angles	Students Enlist the types of alternate angles.	What are alternate angles? List out the types.
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Alternate Interior Angle

<u>Alternate Interior Angle</u>	Alternate Interior Angles are the pair of angles that lie between (inside) the two lines and are on opposite sides of the transversal.	Students describe the alternate interior angle.	How do you describe alternate interior angle?
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These angles are formed inside the parallel lines. They sit in alternate (zig-zag) position across the transversal.

content Learning Experience
Example



Here the angles $\angle 3$ and $\angle 5$ & $\angle 4$ and $\angle 6$ are called alternate interior angle.

Property of Alternate Interior Angle.

1) If a transversal intersect two parallel lines, then each pair of alternate interior angles are equal.

In above figure $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$.

2) If a pair of alternate interior angles is equal then the two lines are equal.

Equality of alternate Interior angle is a test for parallelism.

Behavioural objectives

Evaluation

Students identify pairs of alternate interior angle.

Students state the property of alternate interior angles.

Property of Alternate Interior Angle

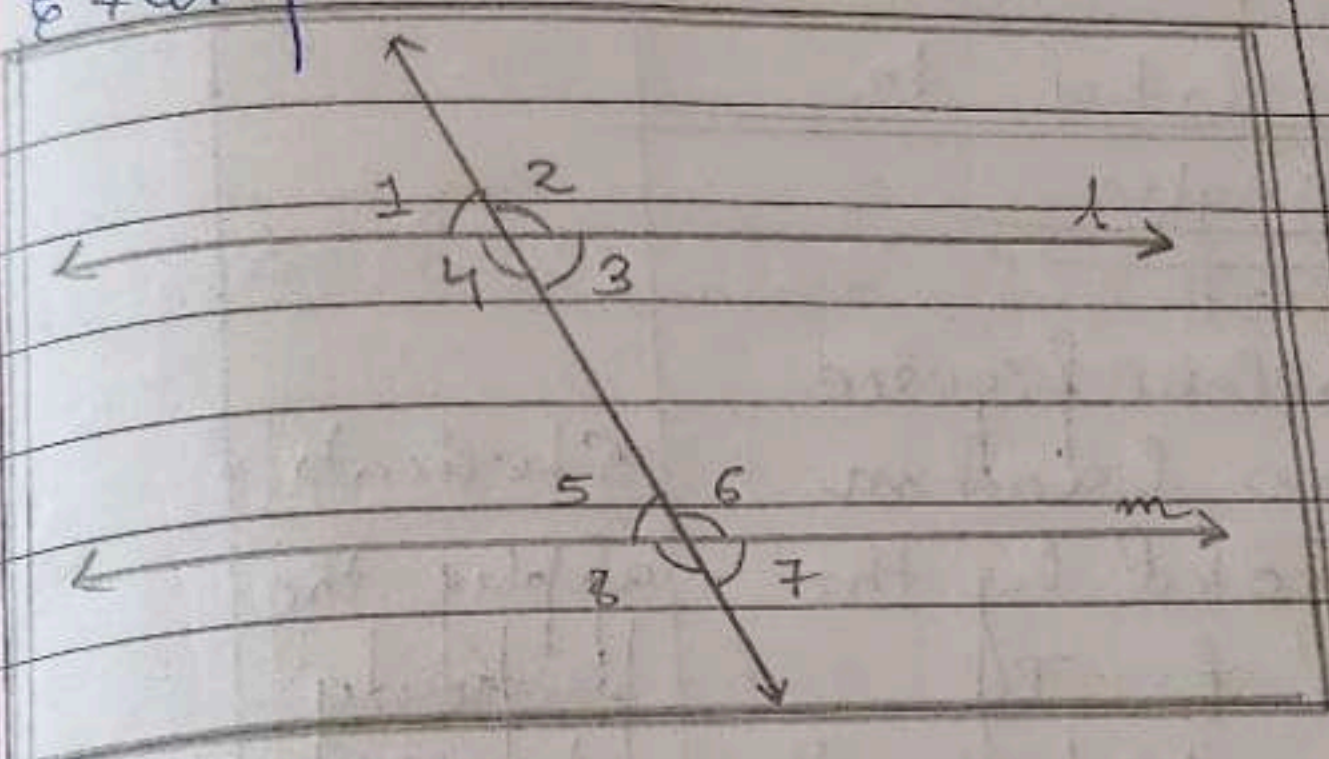
Learning	Behavioural objectives	Evaluation
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Alternate Exterior Angles

Alternate exterior angles are the pairs of angles that lies outside the two lines and on opposite sides of the transversal.

These angles are formed outside the parallel lines. They also lie in alternate position across the transversal.

Example:



Here the angles $\angle 1$ and $\angle 7$, and $\angle 2$ and $\angle 8$ are called alternate exterior angle.

Property of Alternate Exterior Angles

- If a transversal intersect two parallel lines, each pairs of alternate exterior angle are equal.

Learning objectives

In the above figure $\angle 1 = \angle 2$ and $\angle 3 = \angle 4$

2) If a pair of alternate exterior angles is equal then the two lines are parallel

Equality of alternate exterior angles is also a test to check whether lines are parallel.

Behavioural objectives

Students state the property of alternate exterior angles

Evaluation

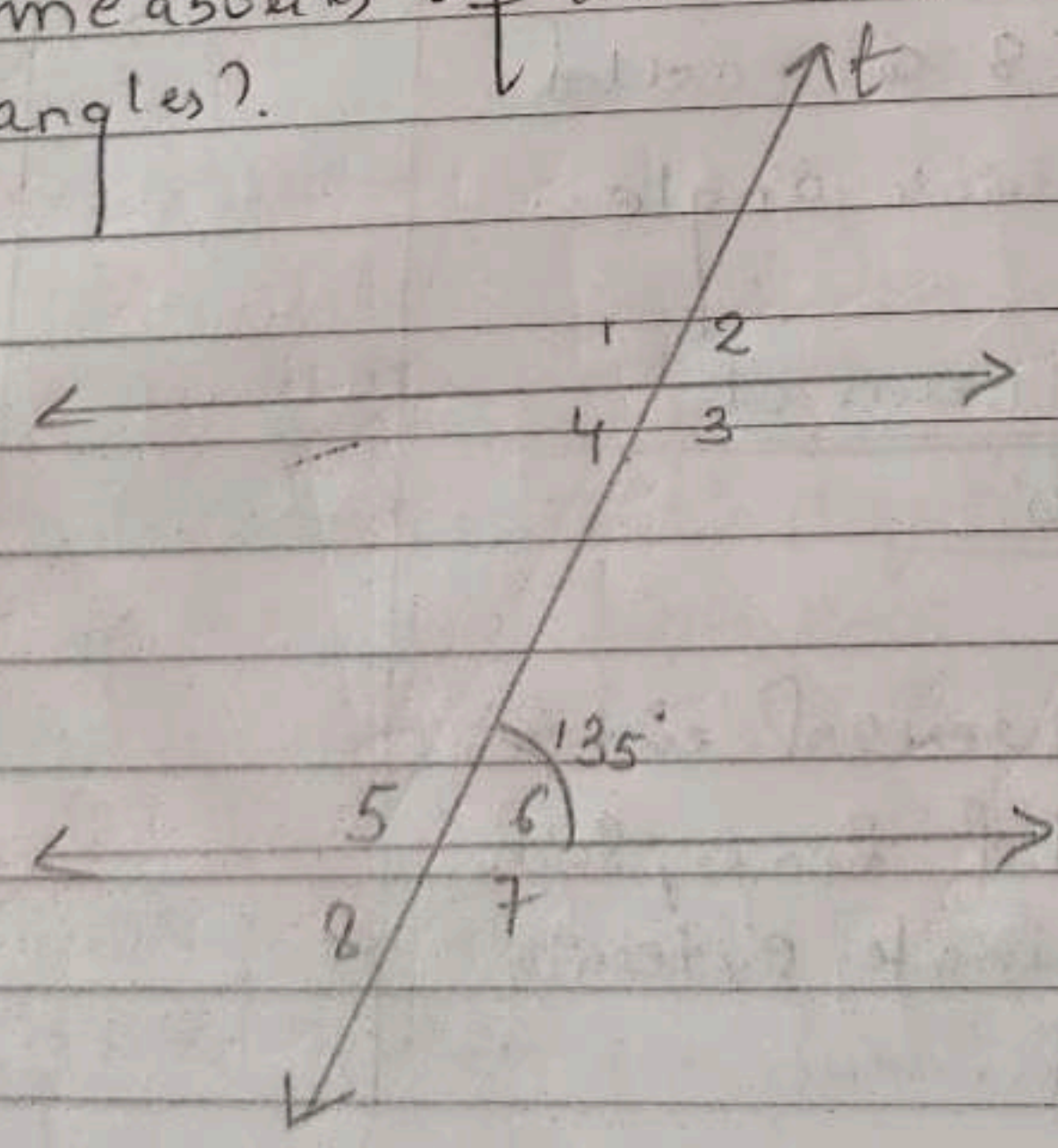
State the property of alternate exterior angles

Problem related to Alternate angles

Problem related to Alternate angles

1) In the below figure parallel lines l and m are intersected by the transversal t. If $\angle 6$ is 135° , what are the measures of the other angles?

Students apply the property of alternate angles to solve simple numerical problem.



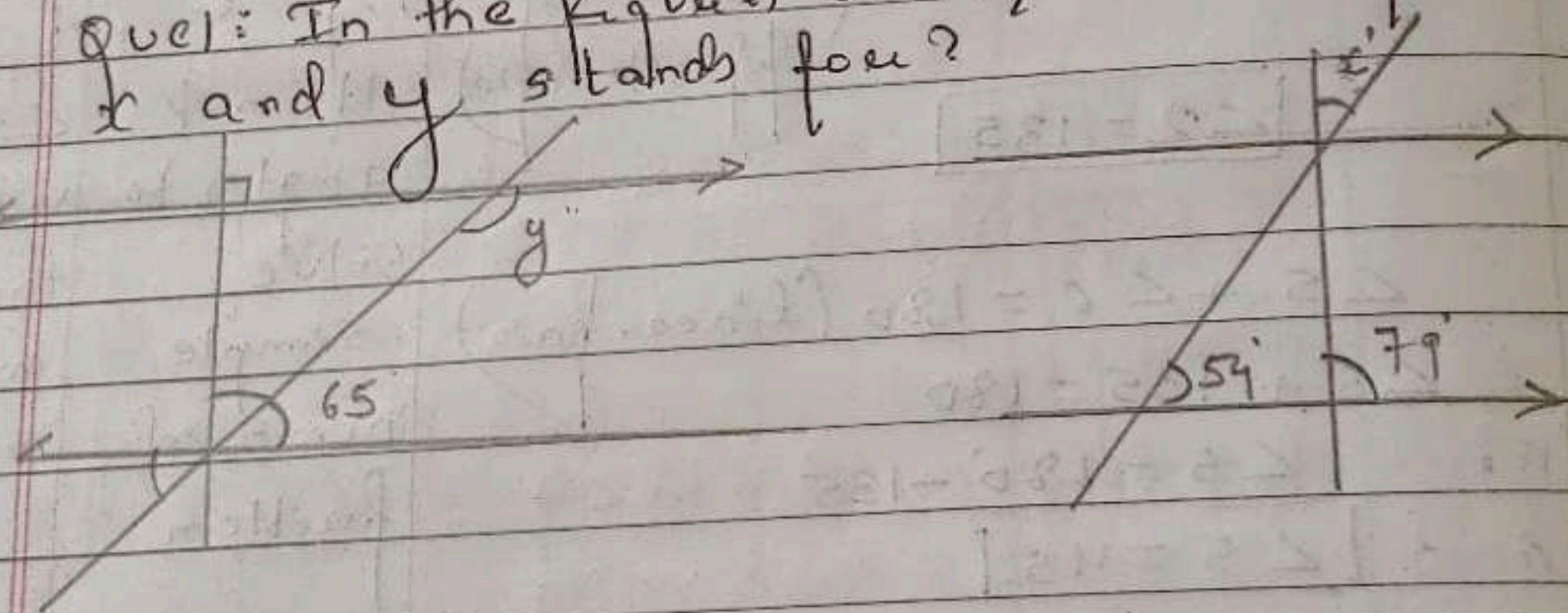
Learning Experiences	Behavioural Objectives	Evaluation
sol) Given $\angle 6 = 135^\circ$ Now, according to property of corresponding angle $\angle 2 = \angle 6 = 135^\circ$ $\angle 2 = 135^\circ$ Now, according to the property of vertically opposite angle $\angle 8 = \angle 6 = 135^\circ$ $\angle 8 = 135^\circ$ $\Rightarrow \angle 4 = \angle 8 = 135^\circ$ (Corresponding angle) $\angle 4 = 135^\circ$ $\angle 2 = \angle 4 = 135^\circ$ (vertically opposite angle) $\angle 2 = 135^\circ$ $\angle 5 + \angle 6 = 180^\circ$ (linear pair) $\angle 5 + 135 = 180$ $\angle 5 = 180 - 135$ $\angle 5 = 45^\circ$ Similarly $\angle 7 = 45^\circ$ (V.O.A) $\angle 3 = 45^\circ$ (C.A) $\angle 1 = 45^\circ$ (V.O.A)	Students apply the property of alternate angles to solve the simple numerical problem.	In the below figure parallel line l and m are intersected by the transversal t. If $\angle 3$ is 50 what is the measure of $\angle 6$?

VII SUMMARY

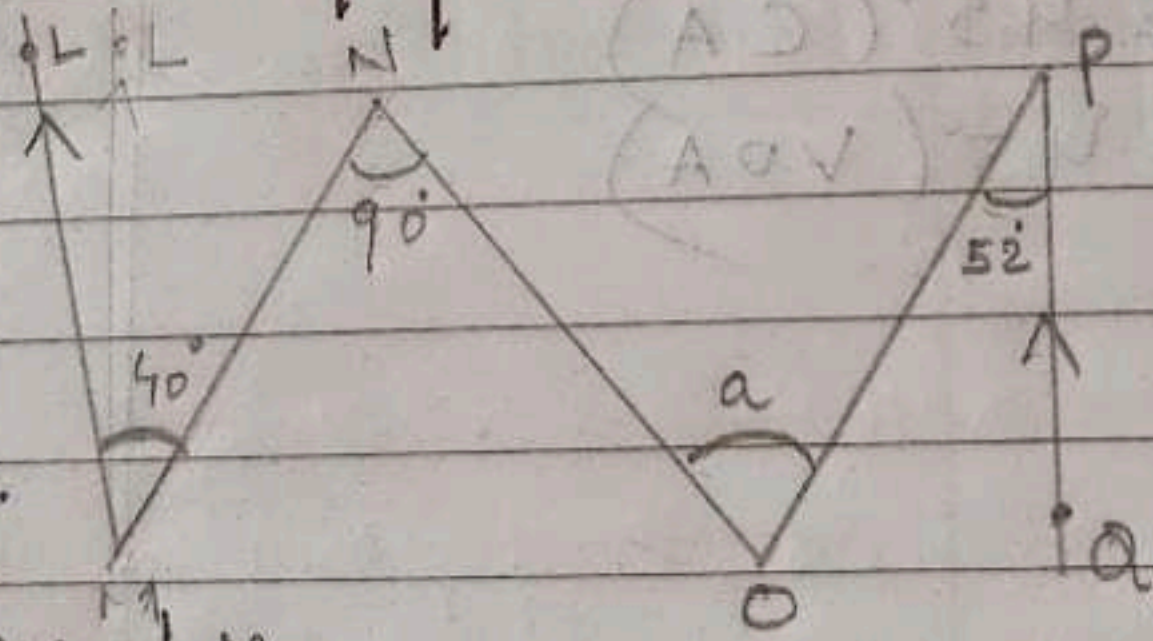
- Alternate angle lie opposite sides of the transversal. There are two types of alternate angles:
 - a) Alternate interior angles are the angles that lie inside the transversal
 - b) Alternate exterior angles are the angles that lie outside the transversal
- Property:
 - a) If a transversal intersect two parallel lines, each pair of alternate angles are equal
 - b) If pair of alternate angles are equal then the two lines are parallel.

VIII EVALUATION

Ques 1: In the figure below, what angles do x and y stand for?



Ques 2: what is the measure of angle $\angle NOP$ in below figure?



Signature of the Teacher Terence

Signature of the Guide Teacher

LESSON PLAN-21

I

PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. A. Karsha
Name of the Practising School : GSS South Point
Date :
Period : IV
class : VII
Subject : Mathematics
Name of the Lesson : A tale of three intersecting lines
Name of the Topic : Equilateral triangle
Duration of the Period : 40 mins

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class students will be able to:

- i) define Equilateral triangle.
- ii) state the properties of an Equilateral triangle.
- iii) explain the steps of constructing an Equilateral triangle when side is given.
- iv) construct an Equilateral triangle when one side is given.

III TEACHING AID

Use of flashcards to demonstrate the steps of constructing an Equilateral triangle.

IV ENTRY BEHAVIOUR

que 1: What is a triangle?

Ans. A closed figure with three sides and three angles.

que 2: How many vertices does a triangle have?

Ans. Three vertices.

que 3: Give some examples of triangle you observe in your daily life?

Ans. Traffic sign board, slice of pizza.

que 4: How many angles does a triangle have?

Ans. Three angles.

que 5: Do you know a triangle which have equal measure of sides and equal measure of angle?

Ans. Yes!

V ANNOUNCEMENT OF THE TOPIC

So, student today we are going to study about 'Equilateral triangle'.

VI INSTRUC

content: Learning

Equila-
teral
triangle

Equilateral

An Equilateral triangle has all three sides equal.

The three angles are also equal.

Each angle is 60°.

So, an

equilateral triangle has all three sides equal and all three angles are equal.

angle

angle

angle

Definition of an Equilateral triangle

Properties

of an Equilateral triangle

1. All three sides are equal.

2. All three angles are equal.

3. Each angle is 60°.

4. The sum of all three angles is 180°.

5. The triangle is equilateral.

6. The triangle is equilateral.

7. The triangle is equilateral.

8. The triangle is equilateral.

9. The triangle is equilateral.

10. The triangle is equilateral.

11. The triangle is equilateral.

12. The triangle is equilateral.

13. The triangle is equilateral.

14. The triangle is equilateral.

15. The triangle is equilateral.

16. The triangle is equilateral.

17. The triangle is equilateral.

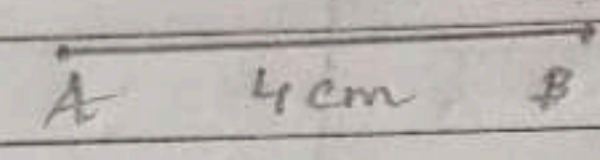
18. The triangle is equilateral.

19. The triangle is equilateral.

20. The triangle is equilateral.

INSTRUCTIONAL PROCEDURE

Learning Experiences	Behavioural objectives	Evaluation
<u>Equilateral Triangle</u> An Equilateral triangle in which all three sides are equal in length. The term Equilateral comes from two words: • Equi means Equal • Lateral means sides So, an Equilateral triangle having all its sides equal. Because all sides are equal, all the three angles are also equal and each angle measures 60°.	Student defines an Equilateral triangle	Define an Equilateral triangle.
<u>Properties of an Equilateral triangle.</u> Property 1: All three sides are equal. In an Equilateral triangle, the length of all its sides are equal. This is the main property that makes the triangle 'Equilateral'. Property 2: All three angles are equal. Since all sides are equal, all the angles of the triangle are equal.	Students state the properties of an Equilateral triangle.	State the properties of an Equilateral triangle.

content	Learning Experiences	Behavioural objectives	Evaluation
construction of an isosceles triangle of side 4cm	Let us consider an Equilateral triangle of side 4cm. Now for constructing an Equilateral triangle we can't straightly use the ruler but the triangle will be not perfect. So, for the precise construction we can take help of the compass.	Students Explain the steps of constructing an Equilateral Δ where side is given	
	Step 1: Make a straight line of measure 4cm using a ruler and a pencil make a straight line of measure 4cm as shown below		
		Students Construct an Equilateral Δ when one side is given.	
	Step 2: using a compass, construct a sufficiently long arc of radius 4cm from A. Placing your compass on point A and with a radius of 4cm cut an arc.	Explain the steps of constructing an Equilateral triangle where sides are given.	

content Learning

Step 1

Step 2

Step 3

Step 4

Step 5

Step 6

Step 7

Step 8

Step 9

Step 10

Step 11

Step 12

Step 13

Step 14

Step 15

Step 16

Step 17

Step 18

Step 19

Step 20

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Step 24

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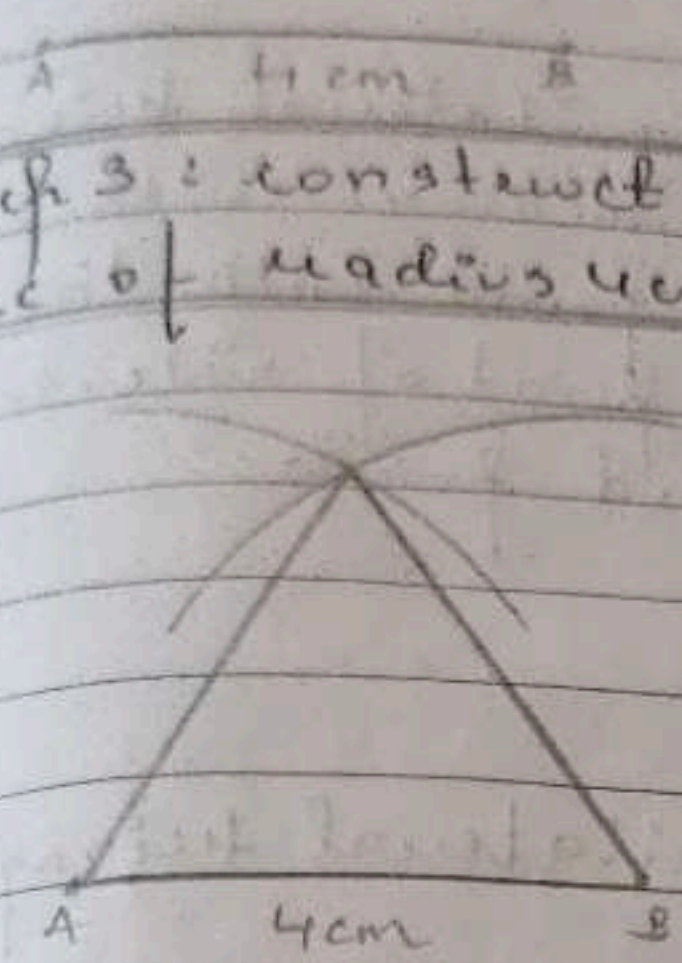
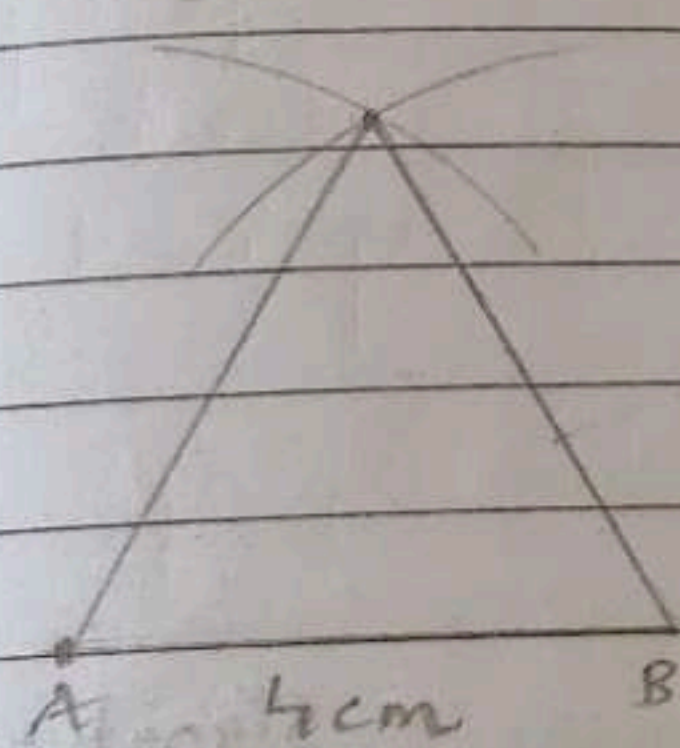
Step 96

Step 97

Step 98

Step 99

Step 100

Content	Learning Experiences	Behavioural Objectives	Evaluation
Step 3: Construct another arc of radius 4cm from B. 	Students construct an Equilateral triangle when one side is given.		
Step 4: Join AC and BC to get the required Equilateral triangle. At least join point A and point B to the point of intersection of two arcs i.e., C. 	Student constructs an Equilateral triangle when one side is given.	Construct a triangle of each 4cm.	

VII SUMMARY

- An Equilateral triangle has all three sides equal.
- All three angles of an Equilateral triangle are also equal.
- For constructing an Equilateral triangle draw a line of given measure with same measure cut an arc from both the ends. Atleast join the point of intersection of both arc to the end points.

VII EVALUATION

que1: what is an Equilateral triangle.

que2: Draw an Equilateral triangle of the following measure:
a) 5cm b) 4cm c) 6cm

que3: How many angles and sides does an Equilateral triangle have?

Ans

Signature of the
Teacher Trainee

Signature of the
Guide Teacher

LESSON PLAN - 22

I. PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. AKANSHA
Name of the Practising School : G. S. South Point
Date :
Period : IV
Class : VII
Subject : Mathematics
Name of the Lesson : A tale of three intersecting lines
Name of the topic : Constructing a triangle when sides are given and triangle inequality
Duration of the Period : 40 mins

II. SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class students will be able to

- i) explain the steps of constructing a triangle when sides are given
- ii) construct a triangle when sides are given
- iii) define triangle inequality
- iv) describe triangle inequality by visualising the construction of circles
- v) apply triangle inequality property to solve simple problems related to triangle.

III. TEACHING AID

use of paper strips of different lengths to demonstrate that only those sets of three

lengths which satisfy the triangle inequality
can form a triangle

IV ENTRY BEHAVIOUR

Ques: How do you describe an Equilateral triangle?

Ans: An Equilateral triangle is a triangle with 3 Equal sides & 3 Equal angles

Ques: What are the main properties of an Equilateral triangle?

Ans: All three sides are Equal and all three angles are Equal

Ques: If I give you one side of an Equilateral triangle will you be able to construct the triangle?

Ans: Yes, we can construct

Ques: If I give you three random lengths, will it always be possible to make a triangle? why or why not?

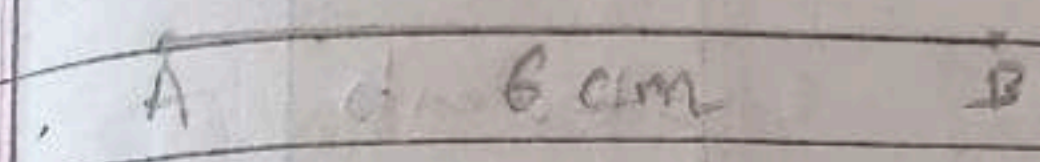
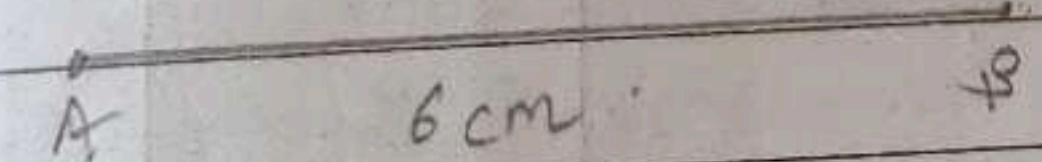
Ans: Silence

V ANNOUNCEMENT OF THE TOPIC

So, students today we are going to study about

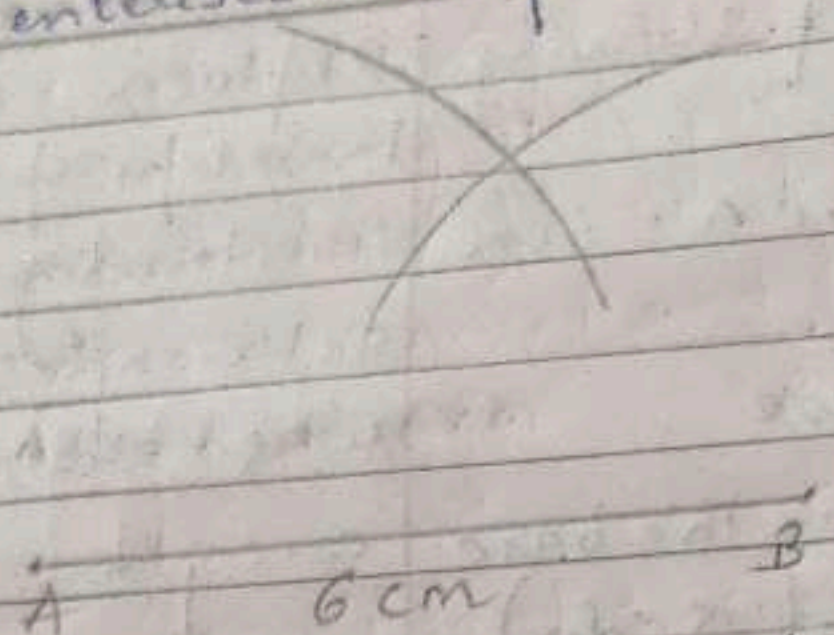
'Constructing a triangle when sides are given and Triangle inequality'

VI INSTRUCTIONAL PROCEDURE

Content	Experiences	Behavioural objectives	Evaluation
Steps of constructing a triangle when sides are given	Suppose we need to construct $\triangle ABC$ with the following measurement $AB = 5\text{cm}$, $BC = 4\text{cm}$, $CA = 6\text{cm}$	Students Explain the steps of constructing a triangle when sides are given	
Step 1: Draw the base	- choose one side as the base (usually the longest side) - Draw $AB = 5\text{cm}$ using scale.		
			Explain the steps of constructing a triangle when sides are given.
Step 2: Set compass for the second side	- Place the compass needle on point A - open the compass to 6cm (length of AC) - Draw an Arc above the line AB	Students construct triangle when sides are given.	
			

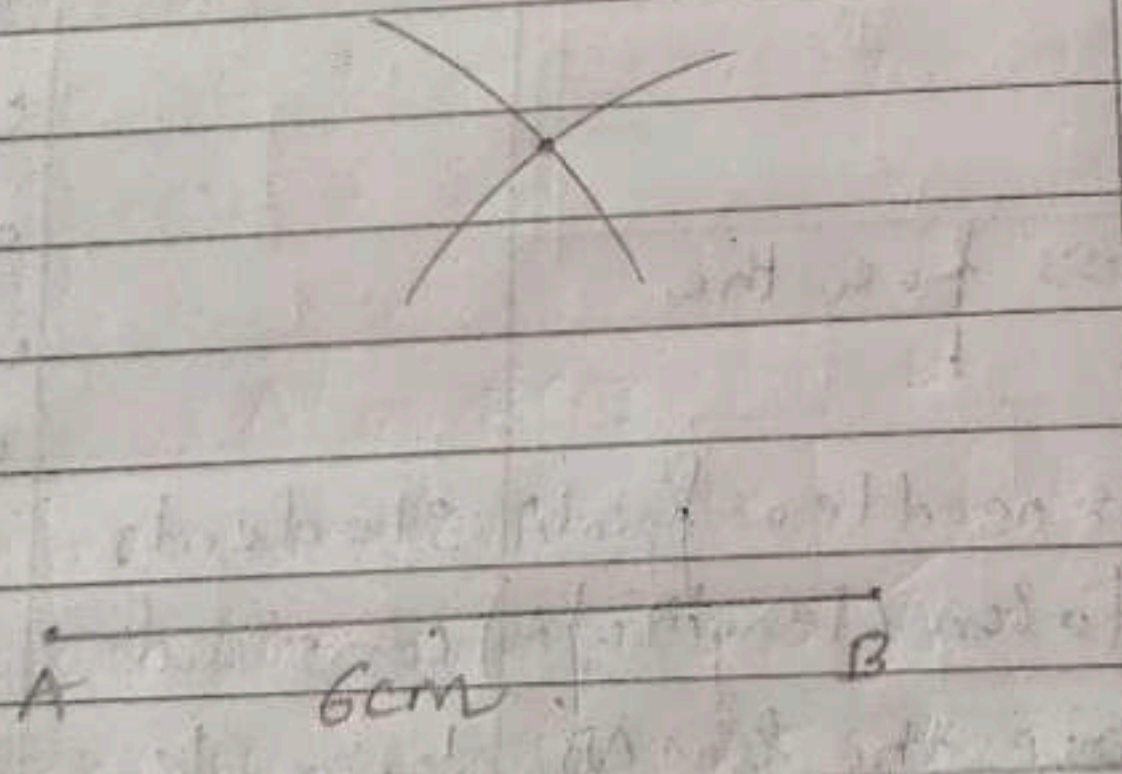
Content Learning Experiences

- Step 3: Set compass for the third side.
- Place the compass needle on point B.
 - Open the compass to 4 cm.
 - Draw another arc so that it intersects the first arc.



Step 4: Mark the intersection point.

Label the intersection point of the arc as C.



- Step 5: Join the points.
- Join AC and BC with a ruler.
 - Now triangle ABC is formed.

Behavioral objectives

Students Explain the steps of constructing a triangle.

Evaluation

Students construct a triangle when sides are given.

Construct a triangle of sides.

5 cm, 6 cm, 4 cm

Content Learning

Triangle inequality

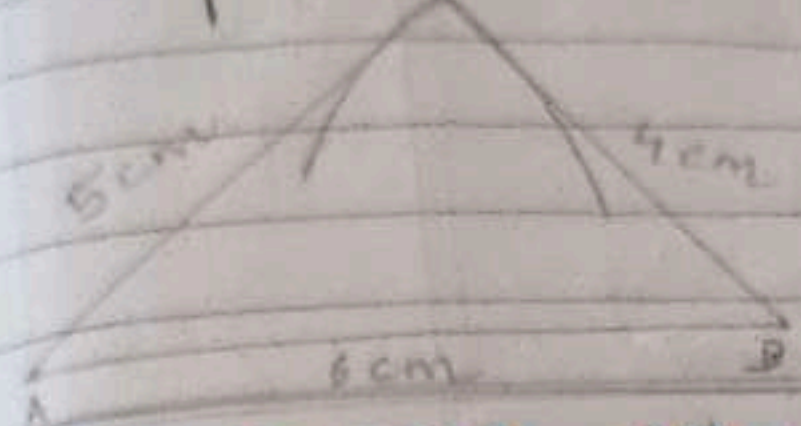
That of a given

If a triangle has sides of lengths 5 cm, 6 cm, and 4 cm, then the triangle is possible.

Learning Experience

Behavioral objectives

Evaluation



Triangle Inequality

Triangle inequality states that the sum of any two sides of a triangle is always greater than the third side.

Students define triangle inequality

If you take any two sides of a triangle and add their length the result will be more than the length of the remaining side.

State the triangle inequality

If this condition is not true for any side then a triangle cannot be formed with those side lengths.

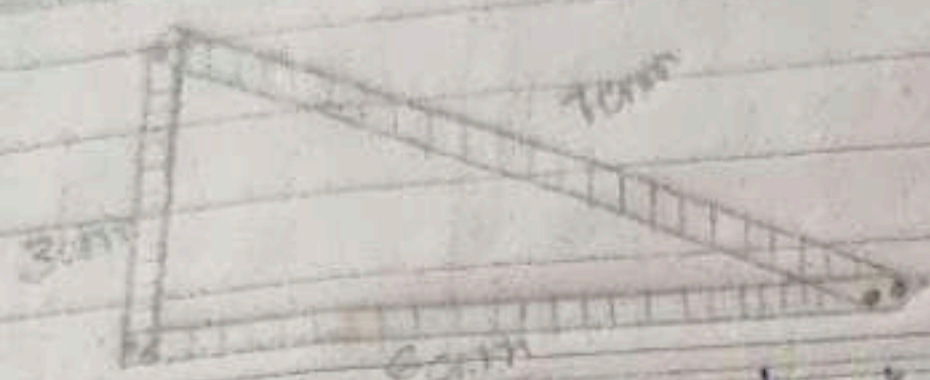
Example:

- For a triangle with sides 5, 6, 7cm
- $5 + 6 > 7$
 - $6 + 7 > 5$
 - $7 + 5 > 6$

Students give example of triangle inequality

All conditions are true → Triangle is possible.

Learning Experiences



Triangle inequality To understand triangle inequality
 visualizing Imagine you are trying to construct a triangle with three
 the given sides using a compass. we use circles (arcs) to check whether
 construction of circles the sides can meet and form a triangle.

Suppose you want to construct $\triangle ABC$ with sides $AB = 5\text{cm}$, $BC = 3\text{cm}$, $CA = 3\text{cm}$.

Step 1: Draw the base $AB = 5\text{cm}$.
 Draw a straight line segment $AB = 5\text{cm}$.

Step 2: Create circle from A.
 • Place the compass at A.
 • Take the radius $AC = 3\text{cm}$.
 • Draw a circle around A. This circle represents all possible locations where point C can be if it is 3cm away from A.

Behavioural objectives

Evaluation

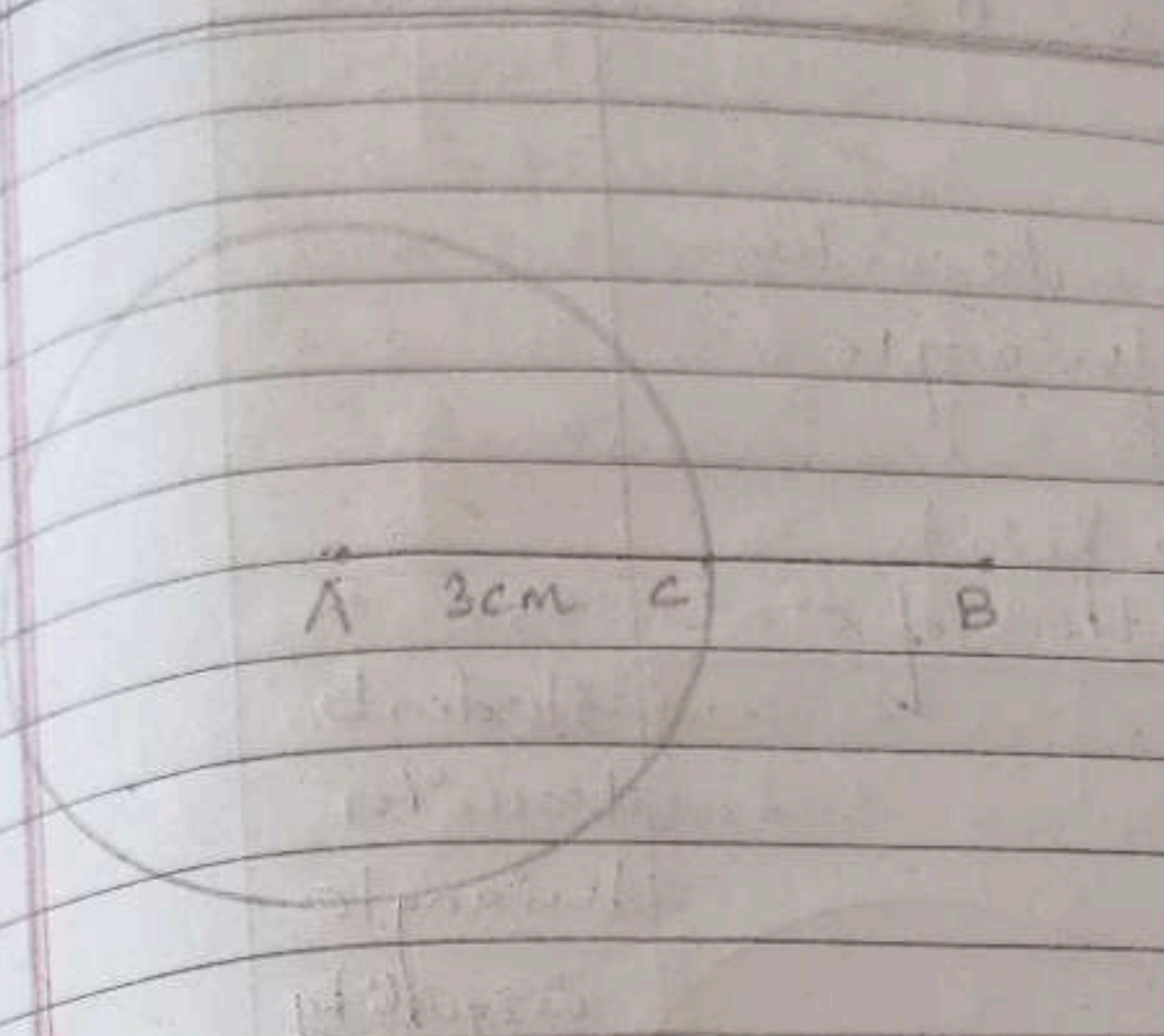
Students describe triangle inequality by visualizing the construction of circles.

Check if triangle is possible for 4cm, 5cm, 6cm.

Learning Experiences

Behavioural objectives

Evaluation

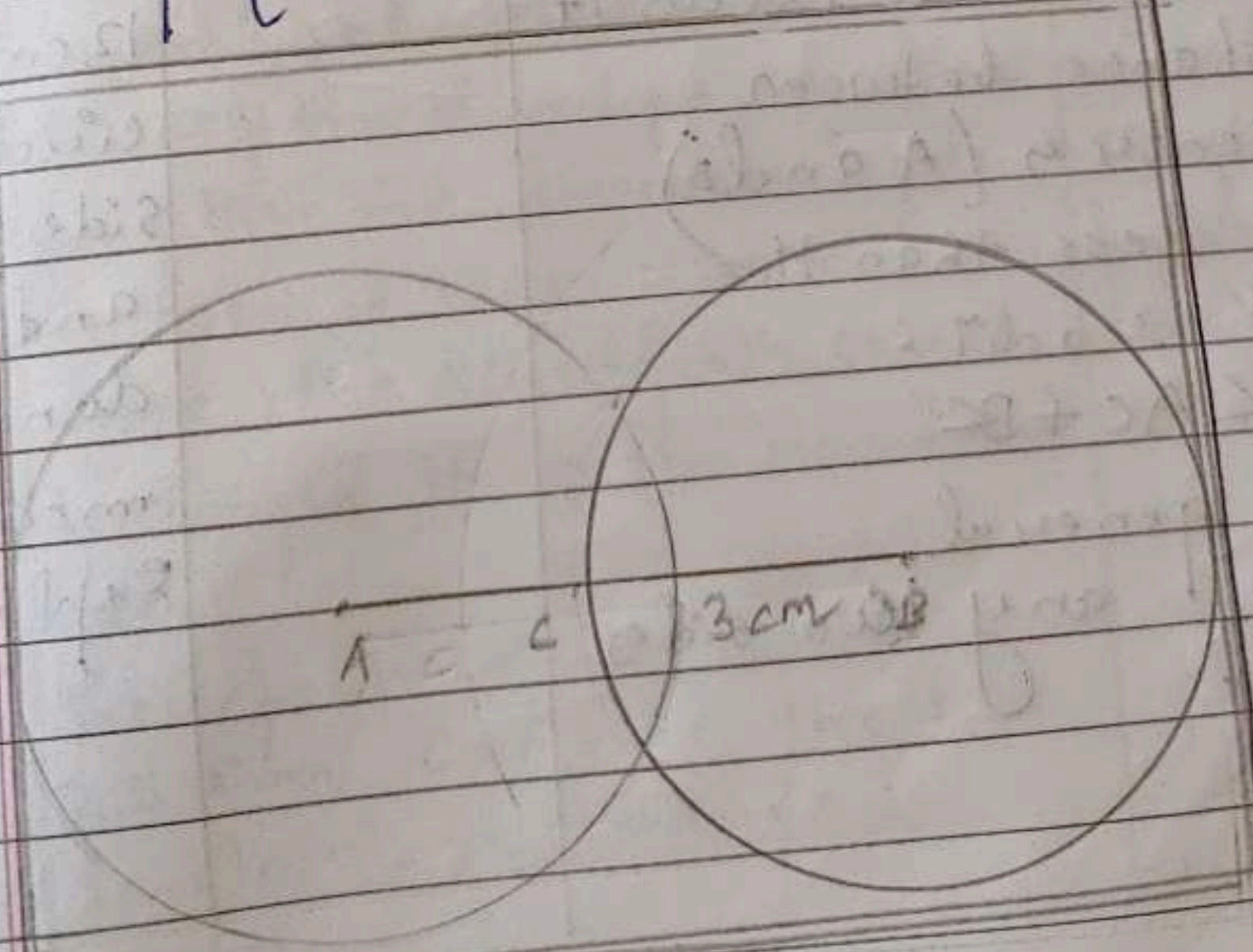


Step 3: Create circle from B

- Place the compass at B.
 - Take the radius $BC = 3\text{cm}$
 - Draw another circle around B
- The circle represent all possible location where point C can be if it is 4cm away from B.

Student describe triangle inequality by visualising the construction of circles

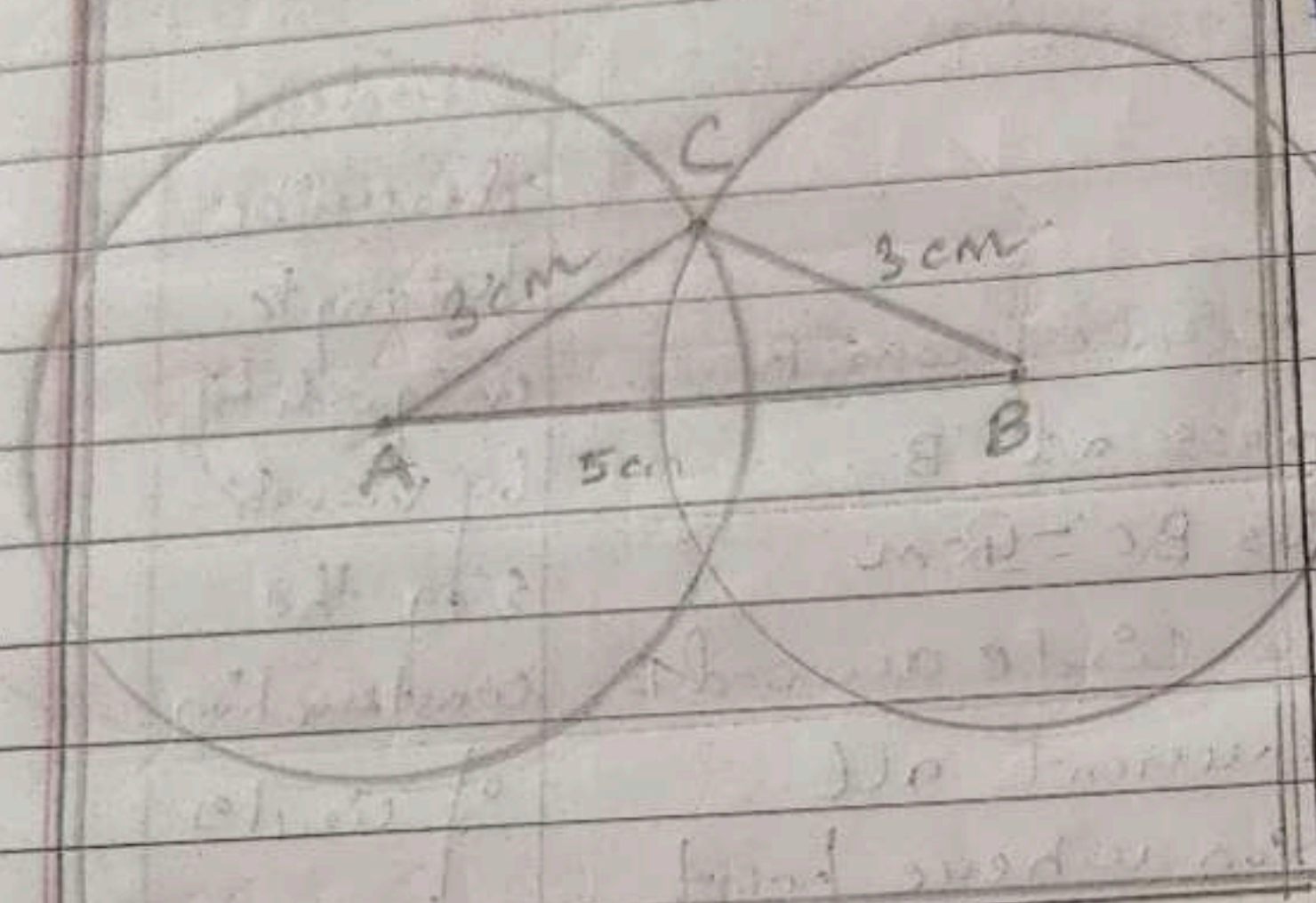
Check if triangle is possible with 4cm, 5cm, 6cm



If the circle do not intersect each other then, it will not form a triangle.

content meaning
Experiences
step 4's observe the intersection

If the two circles intersect then the triangle can be formed.
The intersection point gives the location of C.



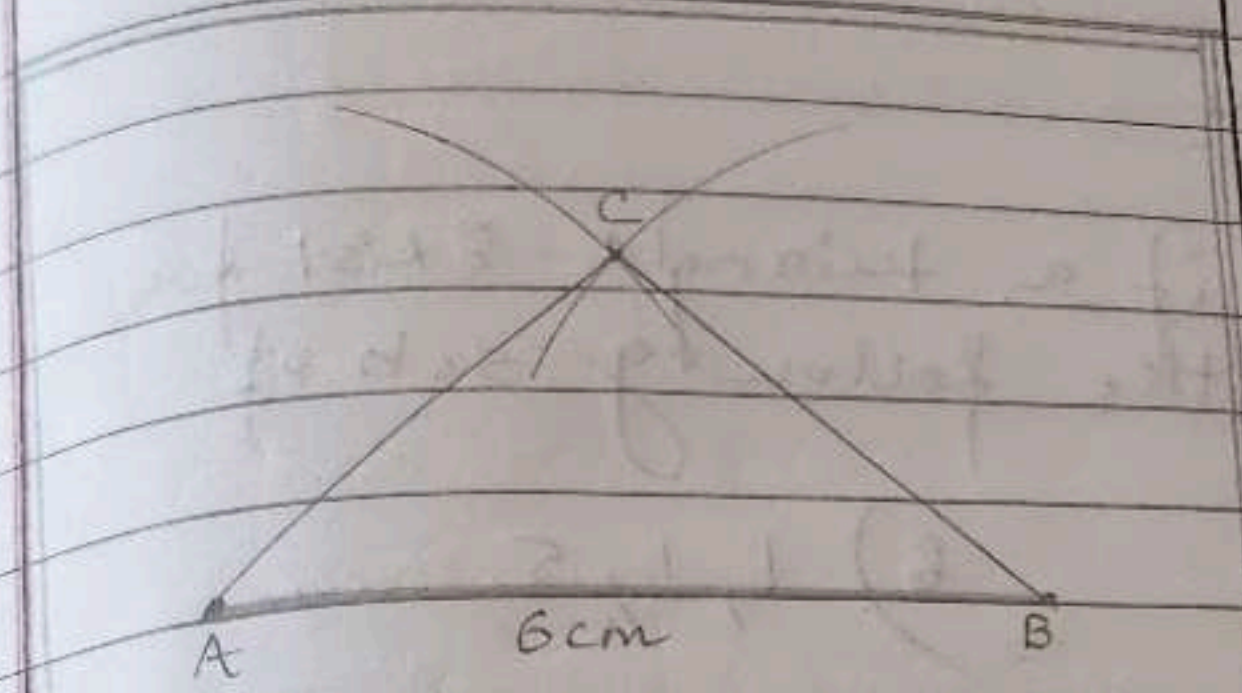
Behavioural objectives
Evaluation

Students describe triangle inequality by visualising the construction of circles

where triangle inequality appears:
For two circles to intersect the distance between their centres (A and B) must be less than the sum of radii:
 $AB < AC + BC$
or in general
→ Sum of any two sides

Can the third side be 12cm if the circles for sides 5cm and 7cm do not meet? Explain.

content meaning
most this will
Problem 1) 40

Learning	Experiences	Behavioural objectives	Evaluation
must be greater than the third side. then the triangle is possible. → If circles does not intersect the triangle will not form.	1) construct a triangle having side lengths 4cm, 4cm, 6cm. 		
2) can we say anything about the existence of a triangle for following sides 10km, 10km, 25km. Sol If the direct path is 25km long then the roundabout path is $10\text{km} + 10\text{km} = 20\text{km}$. Since, the direct path cannot be longer than the roundabout path. Therefore 10km, 10km, 25km can't be the side lengths of a triangle.		student applies triangle inequality property to solve simple problem related to triangle.	By using triangle inequality check whether they form a triangle a) 4, 4, 6 b) 5, 2, 7

VII SUMMARY

- The construction of a triangle when sides is provided involves 5 basic steps:
- Step 1: Draw the base
 - Step 2: Set compass for the second side
 - Step 3: Set compass for third side
 - Step 4: Mark the intersection point
 - Step 5: Join the points
- Triangle inequality states that the sum of any two sides of a triangle is always greater than third side.

VIII EVALUATION

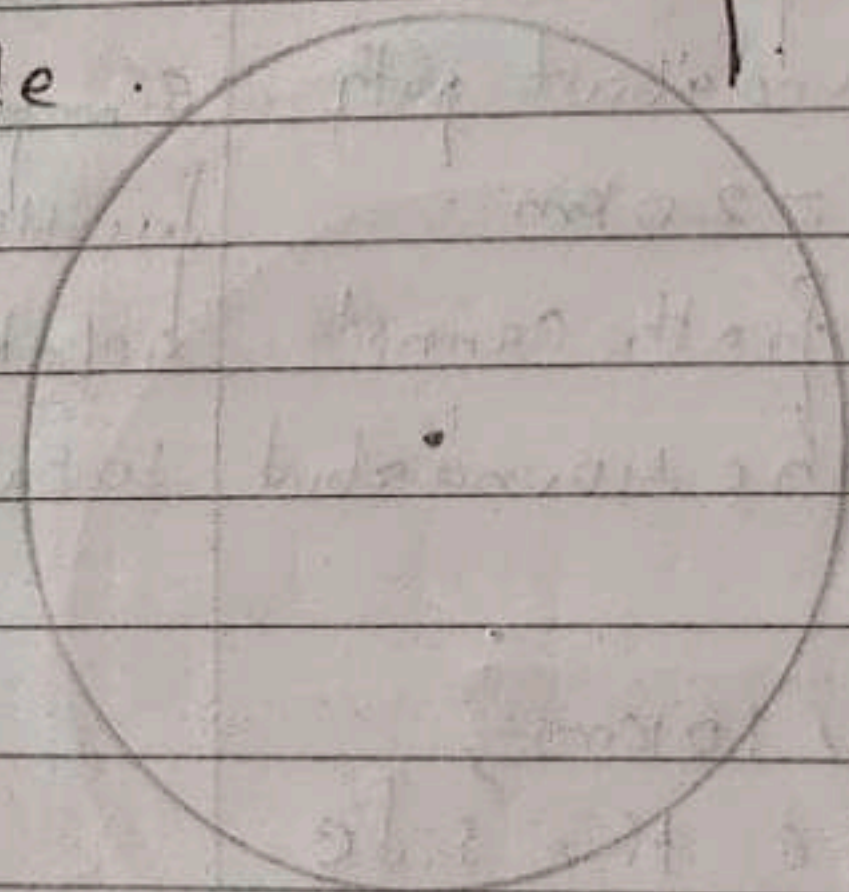
Ques 1: Check if a triangle exist for each of the following set of lengths.

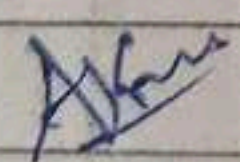
a) 3, 6, 9

b) 1, 1, 5

Ques 2: State the triangle inequality theorem.

Ques 3: Use the point on the circle and the center to form isosceles triangle.




Signature of
the Teacher Trainee

Signature of the
Guide Teacher

I PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Akansha
Name of the Practising School : G. S. South Point
Date :
Period of class : IV
Subject : VII
Name of the Lesson : Mathematics
Name of the topic : A tale of three intersecting lines
Duration of the Period : Construction of triangle when some sides and angles are given
40 mins

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the End of the class students will be able to

- i) explain the steps of construction of triangle when two sides and one angle are given
- ii) construct a triangle when two sides and one angle are given
- iii) describe the steps of construction of triangle when two angles and one side are given
- iv) construct a triangle when two angles and one side are given

III TEACHING AID

use of flashcard to demonstrate the steps of

constructing a triangle when some sides and angles are given

IV ENTRY BEHAVIOUR

Que1: What is triangle inequality?

Ans: The sum of any two sides of a triangle is always greater than the third side.

Que2: If sides of triangle does not satisfy triangle inequality what will happen?

Ans: The triangle will not form.

Que3: Name the instrument we use to measure angles.

Ans: Protractor

Que4: The instrument that helps us to draw arcs and circles.

Ans: Compass

Que5: If two sides and the angle between them are given do you know how to construct triangle?

Ans: Silence

V ANNOUNCEMENT OF THE TOPIC

So, students today we are going to study about

'Construction of triangles when some sides and angles are given'.

VI INSTRUCTION

content learning

Construction when some sides and angles are given

Some triangles we have seen where some sides and angles are given. Some triangles are drawn with some sides and angles given.

Two construction steps are involved in the construction of a triangle when two sides and the included angle are given.

INSTRUCTIONAL PROCEDURE

Meaning
Experiences

Behavioral
Objectives

Evaluation

Construction of triangles
when some sides and angles
are given.

We have seen how to construct triangles when their sidelengths are given. Now, we will take up constructions wherein place of some sidelength, angle measures are given.

Construct $\triangle ABC$ where
 $AB = 5\text{cm}$ $AC = 6\text{cm}$ $\angle A = 60^\circ$

Students
Explain the
Steps of
construction
of triangle
when two
sides and
one angle
are given

Step 1: Draw the base
Draw a line segment $AB = 5\text{cm}$
using a ruler.

Step 2: Construct the given
angle.

At point A, construct $\angle A = 60^\circ$
using a protractor.

Briefly
Explain the
Steps of
construction

Content	Learning Experience	Behavioural objectives	Evaluation
	Step 3: Mark the second side on the angle ray using a compass, take cm/radius and cut an arc on the 60° ray. Label the point where the arc meet the ray as c.	Students explain the steps of construction of triangle when two sides and one angle are given.	
	Step 4: complete the triangle Join Bc with a ruler.	Construct a triangle when one angle and two sides are given a) 8cm, 75°, 7cm. And also explain the steps of construction	

Learning Experiences	Behavioural Objectives	Evaluation
Ques 1) Construct triangle of the measurement where the angle is included between the sides 3cm, 75°, 7cm. Sol ΔPQR is the required triangle Step 1: Draw a line segment of length $PQ = 7\text{cm}$ Step 2: Construct an angle 75° from P. Step 3: Mark the ray at 3cm using compass Step 4: ΔPQR is the required Δ.	Student construct a triangle when two sides and an angle is given.	Construct ΔABC where $AB = 4\text{cm}$ $AC = 5\text{cm}$ $\angle A = 55^\circ$

content	Learning Experiences	Behavioural objectives	Evaluation
Two angle and the included side	Construct ΔXYZ where $YZ = 6\text{cm}$ $\angle Y = 45^\circ$ $\angle Z = 70^\circ$ Step 1: Draw the given side draw the line segment $YZ = 6\text{cm}$ using a ruler Step 2: Construct the first given angle. At point Y, use protractor to draw $\angle Y = 45^\circ$, extend the ray from Y. Step 3: Construct the Second given angle At point Z, use a pro- tractor to draw $\angle Z = 70^\circ$ Extend the ray from Z	Students describe the steps of construction when two angles and one side is given.	what is the first step in angle side angle construction?

Learning Experiences

Behavioural objectives

Evaluation

Step 4: Locate the third vertex external both rays X and Y until they meet. Label the intersection point as X.

Students describes the steps of construction when two angles and one side is given.

1) Construct triangle of the measurement 25° , 3cm , 60°

Students Construct a triangle when two angles and one side is given.

Construct $\triangle XYZ$ where $YZ = 7\text{cm}$, $\angle Y = 40^\circ$, $\angle Z = 65^\circ$

VII SUMMARY

→ The construction of a triangle when two sides and one angle are given involves following steps:

- Step 1: Draw one side using a ruler
- Step 2: Construct the given angle at one end of this side using a protractor
- Step 3: Mark the second side length on the ray using a compass
- Step 4: Join the points to complete the triangle.

→ The construction of a triangle when one side and two angles are given involves basic steps:

- Step 1: Draw the given side
- Step 2: Construct the first angle at one end of the side
- Step 3: Construct the second angle at the other end of the side
- Step 4: Extend both rays until they meet to form the third vertex.

VIII EVALUATION

Que 1: Construct triangles for the following measurement where the angle is included between sides

- a) 3 cm, 75° , 7 cm
- b) 6 cm, 25° , 3 cm

Que 2: Construct triangles for the following measurement

- a) 75° , 5 cm, 75°
- b) 120° , 6 cm, 30°

Signature of the
Teacher Trainee

Signature of the
Guide Teacher

I PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Aransha
Name of the Practising school : Cy 35 South Point
Date :
Period : IV
class : VII
Subject : Mathematics
Name of the Lesson : Tale of three intersecting lines
Name of the topic : Angle sum property of a triangle
Duration of the Period : 40 mins

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class student will be able to

- i) define interior angle of a triangle
- ii) state angle sum property of a triangle
- iii) prove angle sum property of a triangle
- iv) solve numerical problems using the angle sum property.

III TEACHING AID

Use of a working model to visually demonstrate the angle sum property of a triangle.

IV ENTRY BEHAVIOUR

Ques: what is an angle?

Ans: Angle is a figure formed by two rays meeting at a point.

Ques: what do we call an angle that measures exactly 90° ?

Ans: A right angle.

Ques: what is the measure of straight angle?

Ans: 180°

Ques: If we divide a straight angle into two angles, what happens to their sum?

Ans: Their sum is 180° .

Ques: Now look at any triangle, can you tell me what is the sum of its interior angle is?

Ans: Silence.

V ANNOUNCEMENT OF THE TOPIC

So, students today we are going to study about

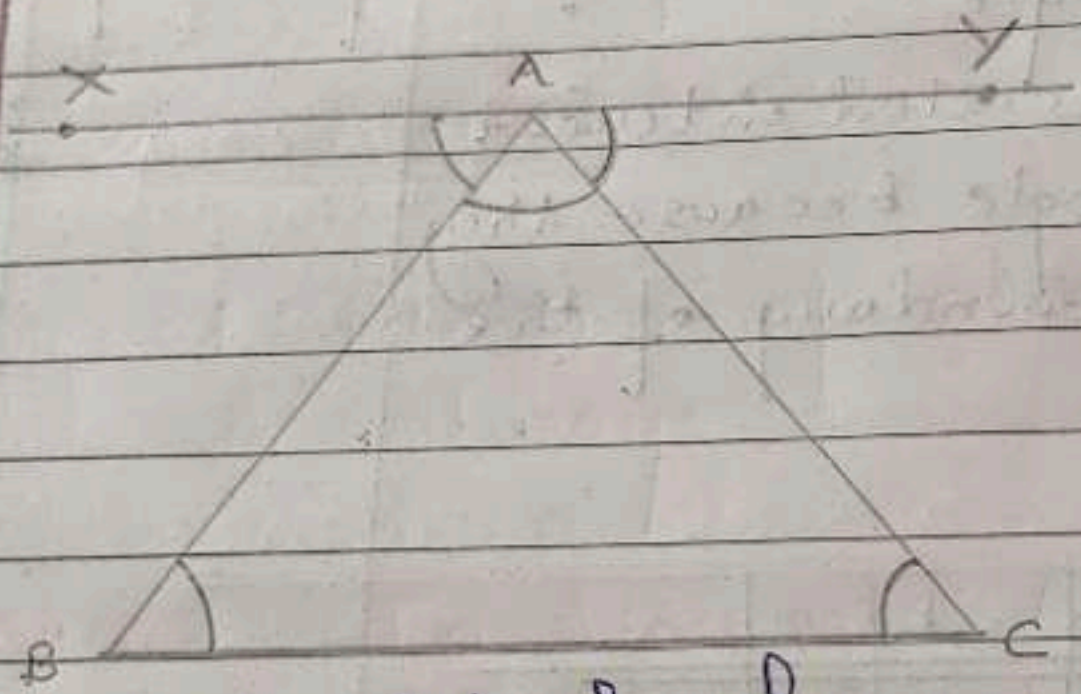
'Angle Sum Property of a triangle'.

VI INSTRUCTIONAL PROCEDURE

Content	Meaning	Experiences	Behavioral objectives	Evaluation
Interior Angle of a triangle	An interior angle of a triangle is the angle formed inside the triangle at each of its vertices. Every triangle has three corners called vertices. At each vertex, two sides meet and form an angle inside the triangle. These angles are called interior angles of a triangle because they lie within the boundary of the triangle.		Students define interior angle of a triangle	Define interior angle of a triangle

Angle Sum Property of a triangle

Angle Sum Property of a triangle	The sum of the three interior angles of any triangle is always 180°. A triangle has three interior angles, one at each vertex. No matter what the shape of triangle is if we add all its three interior angles, the total will always be 180°.		Students state the angle sum property of a triangle.	State angle sum property of a triangle.
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content	Learning Experiences	Behavioural objectives	Evaluation content
Proof of angle sum property of a triangle	<u>Proof of Angle sum Property of a triangle</u> consider a triangle $\triangle ABC$ to find the sum of its angles, we can use the same method of drawing a line parallel to the base/constructed line through A that is parallel to BC.  we need to find $\angle A + \angle B + \angle C$ we know that $\angle B = \angle XAB$ (Alternate Interior angle) $\angle C = \angle YAC$ (Alternate Interior angle) So, $\angle A + \angle B + \angle C = \angle A + \angle XAB + \angle YAC$ $= 180^\circ$ As together they form a straight line. Thus we have proved that sum of the three angles in any triangle is 180°.	student proves angle sum property of a triangle.	Prove that the interior angle of a triangle is equal to 180°.

Journal
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Evaluation

Content

Learning Experiences

Solve using angle sum property of triangle

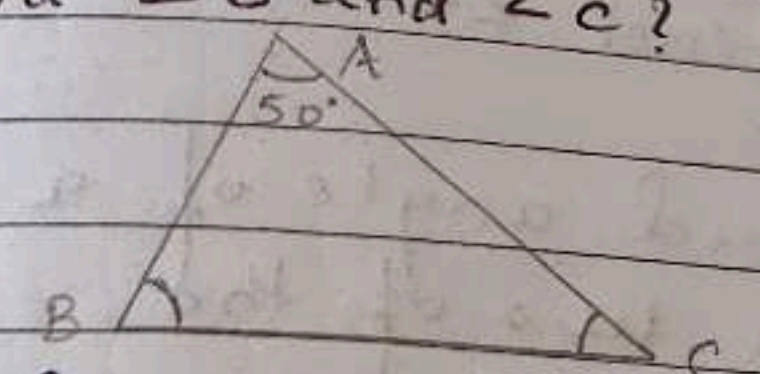
1) Find the third angle of a triangle when two of the angles are 36° , 72° .

Sol As we know that according to angle sum property of a triangle, let the unknown angle be x .

$$\begin{aligned} x + 36^\circ + 72^\circ &= 180^\circ \\ x + 108 &= 180 \\ x &= 180 - 108 \\ \boxed{x = 72^\circ} \end{aligned}$$

The required angle is 72° .

2) Here is a triangle in which we know $\angle B = \angle C$ & $\angle A = 50^\circ$. Find $\angle B$ and $\angle C$?



Sol Given $\angle A = 50^\circ$, $\angle B = \angle C$ let $\angle B = \angle C = x$

According to angle sum property of a triangle.

$$\begin{aligned} x + x + 50^\circ &= 180^\circ \\ 2x + 50 &= 180 \\ 2x &= 130 \\ x &= 130/2 \\ \boxed{x = 65^\circ} \end{aligned}$$

Therefore, $\angle A = 50^\circ$, $\angle B = 65^\circ$, $\angle C = 65^\circ$

Behavioural objectives

Evaluation

Find the third angle when two angles are 30° , 70° .

Students Solves numerical problems using the angle sum property of a triangle.

In a ΔPQR
if $\angle Q = 60^\circ$, $\angle R = 70^\circ$
Find $\angle P$.

Prove that the interior angle of a triangle is equal to 180° .

VII SUMMARY

- A triangle has three interior angles, one at each vertex.
- The interior angles lie inside the boundary of the triangle.
- The angle sum property states that the sum of three interior angles of any triangle is always 180° .
- This property is true for all types of triangles, Scalene, isosceles, Equilateral and right angled.

VIII EVALUATION

Que 1: The sum of angles of a triangle is always _____.

Que 2: In triangle ABC if $\angle A = 40^\circ$ and $\angle B = 90^\circ$, find $\angle C$.

Que 3: Find the third angle of a triangle when two of the angles are $150^\circ, 15^\circ$.

Que 4: A triangle has angles $40^\circ, 60^\circ$ and x° , Find the value of x .

Attns
Signature of the
Teacher Trainee

Signature of the
Lybide Teacher

I PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Akansha
Name of the Practising School : GSS South Point
Date :
Period : IV
Class : VII
Subject : Mathematics
Name of the Lesson : A tale of three intersecting lines
Name of the topic : Exterior angle of a triangle
Duration of the Period : 40 mins.

II SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class students will be able to

- i) define an exterior angle of a triangle
- ii) state the exterior angle property of a triangle
- iii) Explain that the exterior angle is equal to sum of its two interior angles.
- iv) solve numerical problems based on the exterior angle property.

III TEACHING AID

Use of a working model to visually demonstrate the exterior angle property of a triangle.

IV ENTRY BEHAVIOUR

ques: How many angles are there in a triangle?

Ans. Three angles

ques: what is an interior angle of a triangle?

Ans. An angle formed inside the triangle

ques: what is the sum of the three interior angles of a triangle?

Ans. 180°

ques: what happens if you extend one side of a triangle? what do we call it?

Ans. A new angle is formed outside the triangle known as exterior angle.

ques: Do you think this outside angle has any special relationship with the angles inside the triangle?

Ans. Silence

V ANNOUNCEMENT OF THE TOPIC

So, Students today we are going to study about

'Exterior angle of a triangle'.

content

Exterior angle of a triangle

state the exterior angle property of a triangle

VI INSTRUCTIONAL PROCEDURE

Content	Meaning	Experiences	Behavioural objectives	Evaluation
Exterior angle of a triangle	<u>Exterior Angle of a triangle</u> An Exterior angle of a triangle is formed when we extend one of its sides, a new angle is created outside the triangle. when we take any triangle and extend one of its sides, a new angle is created outside the triangle. This angle is called Exterior angle. It lies outside the triangle and is adjacent to one of the interior angles.		Students define Exterior angle of a triangle.	How is an Exterior angle formed?
State the Exterior Angle Property of a Triangle	<u>State the Exterior Angle Property of a Triangle</u> The Exterior angle of a triangle is equal to the sum of the two interior opposite angles. When one side of a Δ is extended, the angle formed outside the triangle is called the Exterior angle. This Exterior angle is always		Students state the property of the Exterior angle of a triangle.	State the Exterior angle property of a triangle.

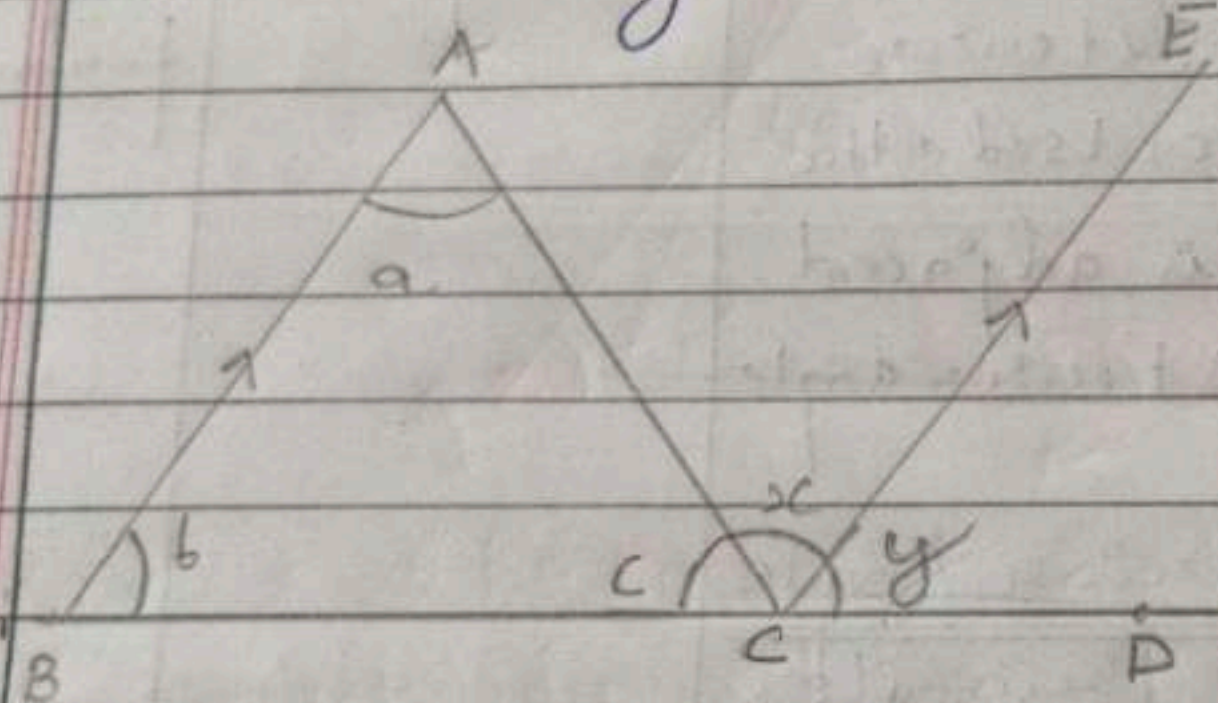
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Learning Experiences

Equal to the sum of the two interior angles that are not adjacent to it.

Proof of Exterior Angle Property

Consider a $\triangle ABC$, where a , b & c are the angles formed. Extend the side BC to D . Now an exterior angle $\angle ACD$ is formed. Draw a line CE parallel to AB . Now x & y are the angles formed, where $\angle ACD = \angle x + \angle y$.



Hence $\angle a = \angle x$ (Pairs of alternate angles)

$\angle b = \angle y$ (Corresponding angle)

Therefore we can say

$$\angle a + \angle b = \angle x + \angle y \quad (1)$$

(From above)

We know that

$$\angle ACD = \angle x + \angle y \quad (2)$$

Substituting (1) in (2)

$$\angle ACD = \angle a + \angle b$$

Hence proved that the exterior angle of a triangle

Behavioural objectives

Evaluation

content

Problem based on

Exterior angle property

Students explain that the exterior angle is equal to the sum of the two opposite interior angles.

Prove that the exterior angle of a triangle is equal to the sum of its opposite interior angles.

Evaluation

Content

Meaning

Experiences
is equal to the sum of two
opposite angles

Behavioural
objectives

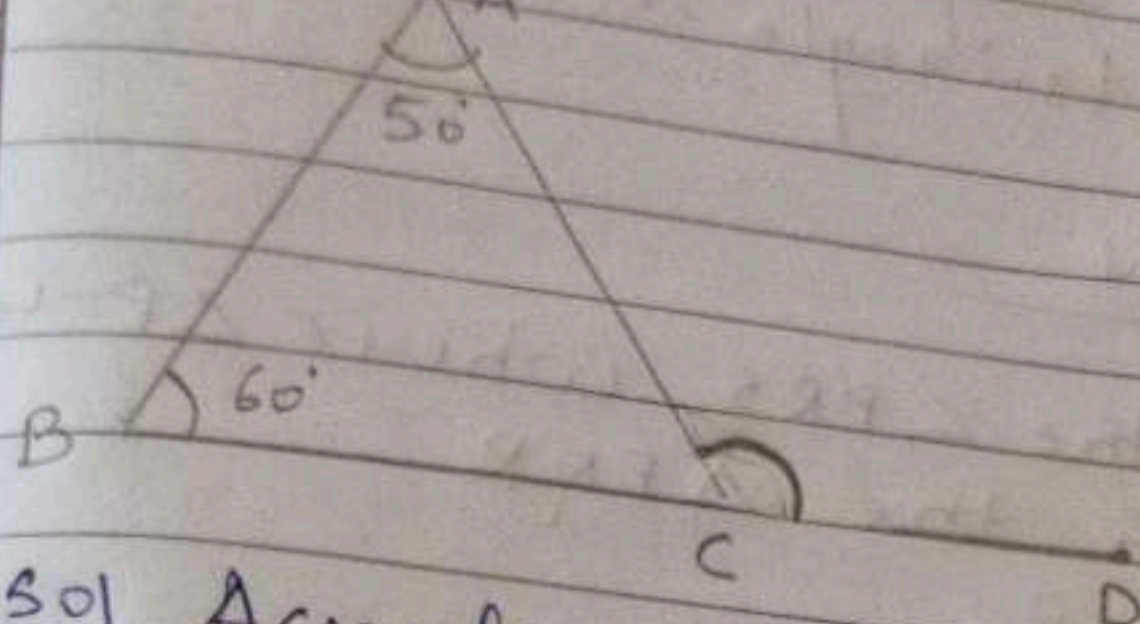
Evaluating

Problem
based
on

Problem Based on Extension
Angle property

Extension
angle
property

1) Find $\angle ACD$ if $\angle A = 50^\circ$
 $\angle B = 60^\circ$



Students
Solves nu-
merical
problems
based on
the extension
angle
property

Sol According to Extension
angle property

$$\angle A + \angle B = \angle C$$

$$50^\circ + 60^\circ = \angle C$$

$$\boxed{\angle C = 110^\circ}$$

In $\triangle ABC$

$$\angle A = 35^\circ$$

$$\text{and } \angle B =$$

$$48^\circ. \text{ Find}$$

the ext-

erior

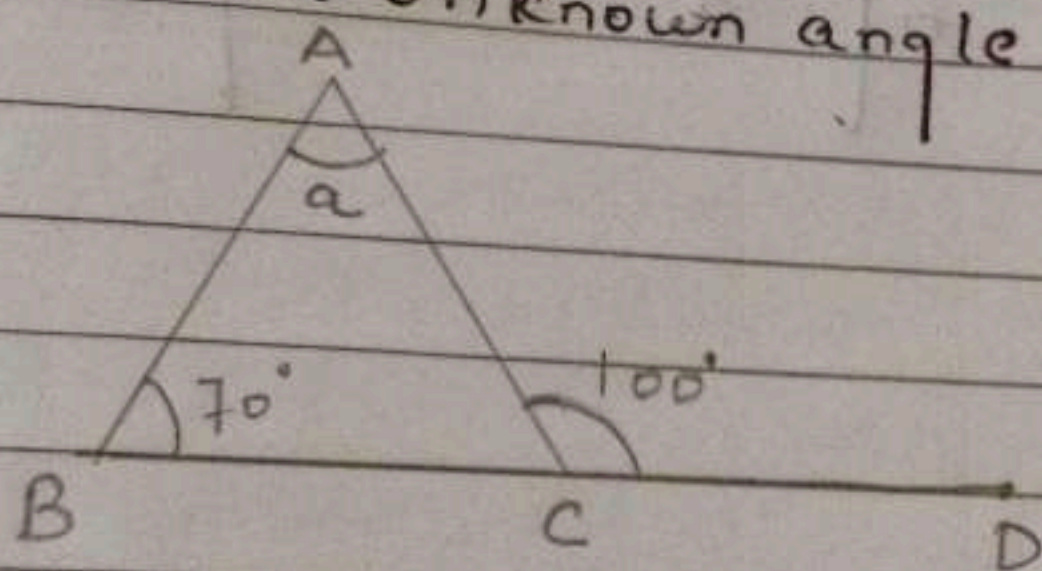
angle

at

vertex

C.

2) Find the unknown angle



Sol: According to Extension
angle property

$$\angle a + 70^\circ = 100^\circ$$

$$\angle a = 100 - 70$$

$$\boxed{\angle a = 30^\circ}$$

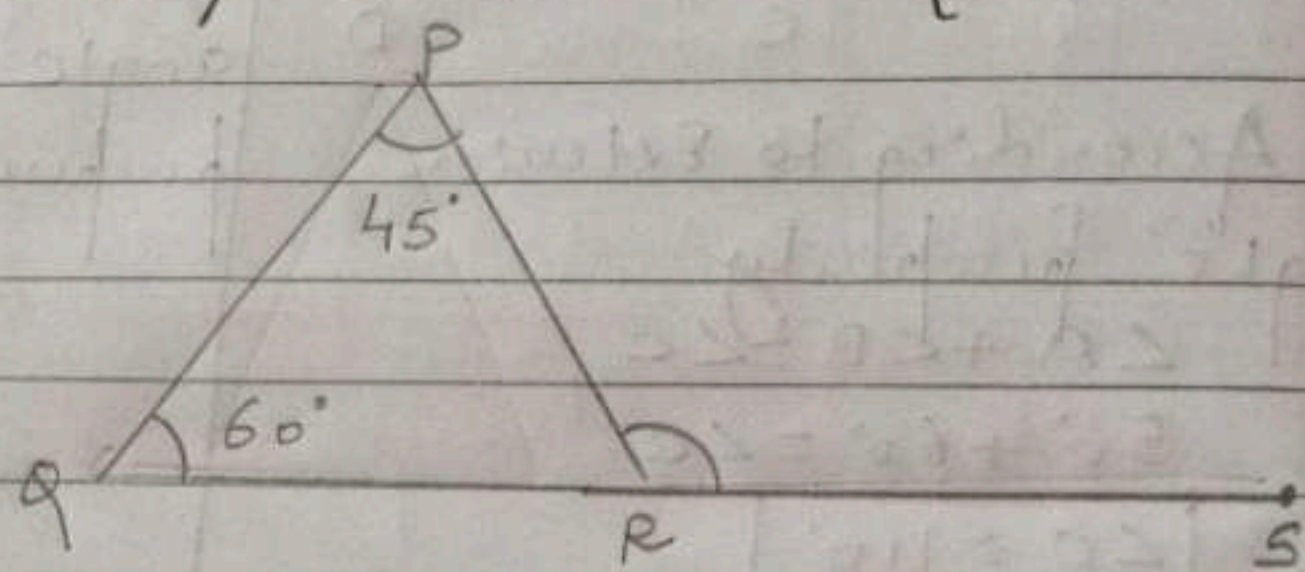
The unknown angle is 30°

VII SUMMARY

- when one side of a triangle is extended, it forms an exterior angle.
- Exterior angle property states that the exterior angle of a triangle is equal to the sum of the two interior opposite angles.
- This property helps in finding unknown angles in a triangle.

VIII EVALUATION

Ques 1: Find the $\angle PRS$, where $\angle P = 45^\circ$, $\angle Q = 60^\circ$, in the $\triangle PQR$



Ques 2: State the Exterior angle property of a triangle.

Aksh

Signature of the
Teacher Trainee

Signature of
the Guide
Teacher

I. PRELIMINARY INFORMATION

Name of the Teacher Trainee : S. Akansha
Name of the Practising School : GSS South Point
Date :
Period :
Class : IV
Subject : VII
Name of the Person : Mathematics
Name of the topic : A tale of three intersecting lines
Duration of the Period : Construction related to Altitudes of triangles
: 40 mins

II. SPECIFIC INSTRUCTIONAL OBJECTIVES

At the end of the class, students will be able to:

- i) define the term altitude of a triangle.
- ii) construct an altitude from a given vertex to its opposite side using a set square and ruler.
- iii) distinguish between the height and side of a triangle when constructing altitudes.
- iv) justify that the altitude of a triangle is always perpendicular to the base.